Journal of Econometrics 216 (2020) 430–449

Contents lists available at ScienceDirect

Journal of Econometrics

journal homepage: www.elsevier.com/locate/jeconom

Option market trading activity and the estimation of the pricing kernel: A Bayesian approach

Giovanni Barone-Adesi [a], Nicola Fusari [b], Antonietta Mira [c,d], Carlo Sala [e,*]

[a] Swiss Finance Institute at Università della Svizzera Italiana (USI), Institute of Finance (IFIN), Via G. Buffi 13, CH-6900 Lugano, Switzerland
[b] The Johns Hopkins University Carey Business School, Baltimore, MD 21202, United States of America
[c] Università della Svizzera Italiana (USI), Data Science Lab, Institute of Computational Science, Via G. Buffi 13, CH-6900 Lugano, Switzerland
[d] University of Insubria, Como, Italy
[e] ESADE Business School, University Ramon Llull, Department of Economics, Finance and Accounting, Avenida de Torreblanca 59, 08172, Barcelona, Spain

ARTICLE INFO

Article history:
Received 23 June 2018
Received in revised form 22 October 2019
Accepted 4 November 2019
Available online 23 November 2019

JEL classification:
G10
G13
G14
G17

Keywords:
Pricing kernel
Pricing kernel puzzle
Physical measure
Dirichlet process
Bayesian nonparametric estimation
Options
S&P 500 index

ABSTRACT

We propose a nonparametric Bayesian approach for the estimation of the pricing kernel. Historical stock returns and option market data are combined through the Dirichlet Process (DP) to construct an *option-adjusted* physical measure. The precision parameter of the DP process is calibrated to the amount of trading activity in deep-out-of-the-money options. We use the option-adjusted physical measure to construct an option-adjusted pricing kernel. An empirical investigation on the S&P 500 Index from 2002 to 2015 shows that the option-adjusted pricing kernel is consistently monotonically decreasing, regardless of the level of volatility, thus providing an explanation to the well known U-shaped pricing kernel puzzle.

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1. Introduction

The pricing kernel (PK), or state price per unit probability, contains information about investors' risk preferences and beliefs, and it is the basis of every asset pricing model. In a simple one-period representative agent economy, with no frictions and concave utility function, the PK is inextricably linked to the distribution of future returns under the physical (or statistical) and the risk-neutral (or pricing) measures (see Rubinstein (1994)). Specifically, the PK at time t, over the horizon $T = t + \tau$ (with $\tau > 0$) is defined as:

$$M_{t,T}(S_T) = e^{-r_{t,\tau}(T-t)} \frac{q_{t,T}\,(S_T|S_t)}{p_{t,T}\,(S_T|S_t)} > 0, \tag{1}$$

* Corresponding author.
E-mail addresses: giovanni.baroneadesi@usi.ch (G. Barone-Adesi), nicola.fusari@jhu.edu (N. Fusari), antonietta.mira@usi.ch (A. Mira), carlo.sala@esade.edu (C. Sala).

https://doi.org/10.1016/j.jeconom.2019.11.001
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where S_t is the value of the underlying asset at time t, $q_{t,T}$ and $p_{t,T}$ are the risk-neutral and physical conditional probability measures between time t and T, and $r_{t,\tau}$ is the relevant risk-free interest rate over a horizon equal to τ.

Exploiting Equation (1), a large body of literature investigates the shape of the empirical pricing kernel (EPK[1]) by estimating the risk-neutral measure using option market data and the physical measure using historical returns on the underlying asset (see, for example, among many others: Äit-Sahalia and Lo (1998), Jackwerth (2000), Rosenberg and Engle (2002), Barone-Adesi et al. (2008), Liu et al. (2009), Beare and Schmidt (2016), Grith et al. (2017) and references therein). The conditional risk-neutral measure can be estimated relatively easily. On each day, for a fixed tenor, we observe a large amount of options with different strike prices. These cross-sections (or panels) of options contain information about the pricing of future state of the world (represented by the different strike prices) and they allow to infer the shape of the risk-neutral measure over a fixed time horizon. On the contrary, the physical conditional probability measure is much harder to estimate because, at each point in time, we only observe one realization from the return's measure. In order to circumvent this issue, a very common approach is to infer the future conditional physical measure using historical returns. This approach is thus by construction *backward-looking*, and it is, in general, affected by the length/amount of past data used in the estimation. As a consequence, for the estimation of the EPK, there is a mismatch between a naturally *forward-looking* risk-neutral measure (the numerator of the EPK) and a *backward-looking* physical measure (the denominator of the EPK). Taking the ratio of the two possibly informationally *misaligned* measures is theoretically inconsistent and may lead to ambiguous empirical results.

Under the settings depicted above, the PK is often interpreted as a normalized marginal utility. In presence of non-satiation and no-arbitrage, the PK must be positive and monotonically decreasing in wealth. However, contrary to economic theory, a large body of literature finds, empirically, a non-monotonically decreasing EPK. The most common violations are U- and hump-shaped EPKs. While a U-shaped EPK is characterized by being increasing in the in-the-money region, a hump-shaped EPK shows a concave pattern near at-the-money.[2] These violations, collectively defined as the "empirical pricing kernel puzzle" are inconsistent with the neoclassical definition of investor risk-aversion and market equilibrium.[3]

After the seminal papers of Äit-Sahalia and Lo (1998), Jackwerth (2000) and Rosenberg and Engle (2002), and more than twenty years of follow-up studies, there is still no consensus on the overall shape of the EPK, and the empirical pricing kernel puzzle has emerged as a solid and frequent empirical pattern in many different markets (see Cuesdeanu and Jackwerth (2017) for a comprehensive survey). For example, by looking at the S&P 500 Index, Barone-Adesi et al. (2008) show that a GJR-GARCH model with filtered historical innovations produces monotonically decreasing EPK over the period 2002–2004. However, their model works well for low volatility days — when the statistical properties of the underlying do not change much from day to day, while it still produces U-shaped EPKs in periods of high volatility. Fig. 1 shows the unconditional EPK estimated following Barone-Adesi et al. (2008) over days of high volatility during the 2002–2015 period. High-volatility days are identified as days in which the VIX index is higher that its median.

In this paper we propose a new method to infer the conditional physical measure of returns by merging the information contained in historical returns and in the option market. The stochastic nature of market volatility is what justifies the use of option market data to achieve a better estimation of the physical measures. Throughout the entire dataset, the EPK is usually U-shaped in days with a higher than average volatility, which usually corresponds to a higher than average percentage of deeply-out-of-the-money (DOTM) options listed and traded in the market. During these days, the shape of the risk-neutral measures quickly changes in accordance with new information flowing into the market. It is difficult to capture these sudden changes with an inherently backward-looking physical measure. This leads to a possibly miss-specified physical measure. As a possible solution to this miss-specification, we leverage the fact that there is a link between volatility, the amount of option traded and listed (Mayhew et al. (1999), Wei and Zheng (2010), and references therein) and the forward-looking information embedded in the option market. More precisely, the higher the volatility, the higher is the number of option traded, and the higher is the informational content (French and Roll, 1980; Chakravarty et al., 2004) that can be extracted from options market about the future state of the economy. As in Barone-Adesi et al. (2008) we adopt a GJR-GARCH model augmented with a nonparametric measure of the innovation process to model the dynamics of stock returns. More specifically, we rely on the filtered historical simulation (FHS) of Barone-Adesi et al. (1999) to infer the shape of the conditional return measure (see Section 4.1 for more details of the FHS method). The joint use of the GJR-GARCH model with FHS innovations (GJR-GARCH-FHS) has the ability to capture the extreme returns observed in our sample, thus accounting for time-varying volatility, skewness, and excess kurtosis. Capturing the dynamics of higher moments of the measure is fundamental for the estimation of the EPK (Christoffersen et al., 2013; Chabi-Yo et al., 2007; Chabi-Yo, 2012; Song and Xiu, 2016). We use the GJR-GARCH-FHS to estimate both the physical and risk-neutral measures. We then use a model-free Bayesian method based on the Dirichlet process (Ferguson, 1973) to nonparametrically enrich

1 We denote with EPK the empirical pricing kernel (EPK), which is the estimated pricing kernel (PK).

2 We distinguish between out-of-the-money, at-the-money, and in-the-money regions as regions in which the future value of the underlying is lower, equal to, or larger than its current value.

3 In a complete market framework, projecting a non-monotonically decreasing PK onto generic market returns may lead to equilibrium problems and to the existence of a trading strategy in contingent claims that stochastically dominates the underlying market index (see, for example, Dybvig (1988), Beare (2011), Beare and Schmidt (2016) and Barone-Adesi and Sala (2019)). Under market incompleteness, a non-monotonically decreasing PK also affects option price bounds (Perrakis and Ryan, 1984; Levy, 1985; Constantinides and Perrakis, 2002).

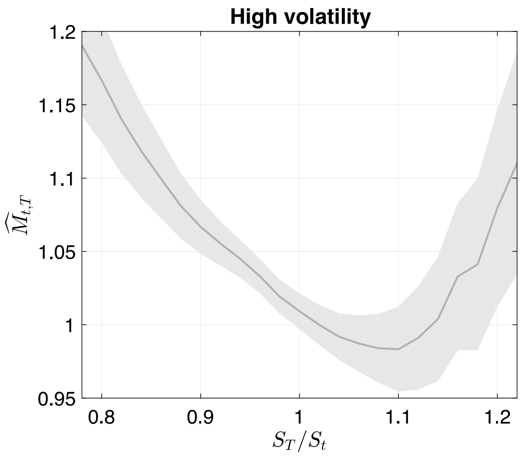

Fig. 1. Empirical pricing kernel (Barone-Adesi et al., 2008): The figure shows the average EPK along with its 90% confidence interval in high volatility days (i.e. VIX index larger than its median value over the sample). The figure is constructed by averaging the daily estimated EPK using a univariate Gaussian kernel with bandwidth selected according to the Silverman's rule of thumb.

the historical physical measure with the information embedded in the option market. The Dirichlet process controls how much information is extracted from the option market and embedded into the construction of the *option-adjusted* physical measure. Given the link between volatility, option trading, and amount of information embedded into option prices, we calibrate the Dirichlet process to the percentage of DOTM options listed in the market.[4] This new option-adjusted physical measure is then used in the estimation of an *option-adjusted* EPK. We test our approach on the S&P 500 Index, on a large sample spanning the period from January 2, 2002 to August 31, 2015. Our empirical results show that the proposed method eliminates the U-shaped EPK puzzle, thus implying economically consistent results in terms of time-varying risk aversions. In order to highlight the model-independent nature of the proposed Bayesian approach, we perform a series of robustness checks. First, we extend the GJR-GARCH-FHS to include a stochastic long-run mean, similar to Engle and Lee (1999) and Christoffersen et al. (2008). This two-factor model is able to better capture the transition from high to low volatility periods. Second, we use a nonparametric approach to estimate the risk-neutral measure, similar to Breeden and Litzenberger (1978) and Figlewski (2008). Regardless of the specific model/approach used to estimate the physical and risk-neutral measures, our empirical findings remain robust.

The estimation of the conditional physical measure using stock returns enriched with option market information is the distinguishing feature of our approach. Previous literature estimates the physical measure by either using only historical observations, thus falling into the mismatch of conditioning information, or by solely using option prices, with the need to rely on dubious parametric assumptions (e.g. Ross (2015)).[5] Our paper is related to Chabi-Yo et al. (2007) and Linn et al. (2017) who also analyze the effect of the informational mismatch in the computation of the probability measures of the EPK. Chabi-Yo et al. (2007), similar to Christoffersen et al. (2013), address the issue by using a model with time-varying volatility in order to capture sudden shifts in the physical measure. However, unless a change in volatility subsumes all changes in investors' expectations, this solution would still lead to informationally inconsistent results. Linn et al. (2017), similar to our approach, rely on option and stock data to recover the physical measure. However, given the econometric structure of their approach, they can only obtain unconditional measures of the EPK.

The remainder of this paper is organized as follows: In Section 2, we define the PK and the issue of the information misalignment in its estimation. In Section 3, we describe the data used in the empirical analysis. Section 4 outlines our new approach to the estimation of the option-adjusted EPK. In Section 5, we present the results of the estimation. Section 6 shows the robustness of the proposed approach to alternative models for the estimation of the physical and risk-neutral measures. Section 7 concludes.

2. Defining the pricing kernel and the empirical pricing kernel

From the fundamental theorems of asset pricing, in presence of a dynamically complete arbitrage-free finite economy, the price of any security is equal to the discounted value of its future expected payoff under the risk-neutral distribution Q or, equivalently, is equal to the discounted value of its future expected payoff times the PK under the physical distribution

[4] Calibrating the Bayesian process to the option volume and the VIX Index produces very similar results.

[5] Ross (2015) shows how, under particular assumptions, it is possible to recover the entire physical measure from a panel of options. However, as shown by Borovicka et al. (2016), most of the assumptions needed are economically difficult to justify.

P (see e.g. Ross (1976)). Both expectations are conditional to the same information set, I_t. It follows that the price of a generic security G_t with single payoff φ_T that depends on the future value of the underlying S_T can be defined as:

$$
\begin{aligned}
G_t &= e^{-r_t\cdot\tau}\mathbb{E}^Q[\varphi_{t,T}(S_T)|I_t] && (2)\\
&= e^{-r_t\cdot\tau}\int_0^\infty \varphi_{t,T}(S_T)q_{t,T}(S_T|S_t)dS_T && (3)\\
&= e^{-r_t\cdot\tau}\int_0^\infty M_{t,T}(S_T)\cdot\varphi_{t,T}(S_T)p_{t,T}(S_T|S_t)dS_T && (4)\\
&= e^{-r_t\cdot\tau}\mathbb{E}^P[M_{t,T}(S_T)\cdot\varphi_{t,T}(S_T)|I_t], && (5)
\end{aligned}
$$

where $\mathbb{E}^Q$ and $\mathbb{E}^P$ are, respectively, the expectations under the risk-neutral $(q_{t,T})$ and physical $(p_{t,T})$ measures and $M_{t,T}(S_T)$ is the PK as defined in Eq. (1). If the S&P 500 Index is used as the underlying asset, the resulting PK is the projected version of the consumption-based marginal rate of substitution.[6] If wealth perfectly correlates with the equity market and investors have a bounded horizon, the *projected* PK has the same pricing implications as the original PK. While a projection might lead to some crucial state variable being missed, as in Ziegler (2007), one should consider our results valid under a representative economy framework. This is the relevant case in our setting, as we rely on historical returns and on European Call and Put options on the S&P 500 Index in our empirical exercise.

In addition to pricing, the PK can be also used to infer investors' risk preferences. From Eq. (1) it is easy to see that, dividing the risk-neutral measure by the risk-adjusted risk-neutral measure (the physical measure), the PK represents a time-adjusted indicator of investors risk preferences. If the utility of the representative investor is a function of wealth only, and in presence of non-satiation, the PK is strictly positive and decreasing. As such, the slope of the PK identifies risk aversion, and lack of monotonicity in the PK implies risk-loving investors' preferences.

It is clear from Eq. (5) that the PK, in its entirety, depends on the information set I_t. While a serious treatment of the role of the information set in the construction of the EPK is *beyond* the scope of this paper,[7] we here provide a brief description of the problem. In a continuous world with no-arbitrage, the PK, $M_{t,T}$, is defined as the Radon–Nikodym derivative of the risk-neutral measure with respect to the physical measure and it is equal to:

$$
\mathrm{PK} := M_{t,T} = \left.\frac{dQ}{dP}\right|_{I_t} > 0 \tag{6}
$$

where P and Q are two σ-finite measures on (Ω, I_t, P) and (Ω, I_t, Q) such that $Q \ll P$ on I_t for all $t \in T$.[8] While the above definition assumes the two measures to be conditional to the same set of information, empirically this is often not the case. The risk-neutral measure is usually estimated from option prices, while the physical measure is commonly estimated using historical returns. The mismatch between an inherently *forward-looking* risk-neutral measure and a *backward-looking* physical measure leads to two different sets of information, with consequences for the estimation of Radon–Nikodym derivative:

$$
\mathrm{PK} := M_{t,T} = \left.\frac{dQ}{dP}\right|_{I_t} \neq \frac{dQ|\mathcal{F}_t}{dP|\mathcal{H}_t} = \widehat{M}_{t,T} := \mathrm{EPK} \tag{7}
$$

with,

$$
\mathcal{F}_t = \sigma\{O^{\mathrm{obs}}_{u,k,\tau} : u \le t, k \in \mathcal{K}_u, \tau \in \mathcal{T}_u\} \qquad \text{and} \qquad \mathcal{H}_t = \sigma\{S_u : u \le t\},
$$

where $O^{\mathrm{obs}}_{u,k,\tau}$ is the observed option price at time u with maturity equal to τ and moneyness k. Moneyness is defined as $k = K/S_t$ where K is the option strike price and S_t is the price of the underlying index at time t. $\mathcal{K}_u$ and $\mathcal{T}_u$ are the set of available strikes and maturity available on day u. A direct consequence of using different information sets ($\mathcal{F}_t$ and $\mathcal{H}_t$) for the estimation of the two measures, and different from what is required by the theory, is that the two information sets might differ substantially.[9] Technicalities aside, Eq. (7) shows the difference between the theoretical PK, $M_{t,T}$, that derives from Eq. (6), and its possibly "biased" empirical counterpart, $\widehat{M}_{t,T}$. Our goal is to show that the use of information embedded in the option market in the estimation of the physical measure can reduce or eliminate such information discrepancy.

6 See Cochrane (2001) for a formal treatment of the theoretical argument, and Äit-Sahalia and Lo (1998), Jackwerth (2000) and Rosenberg and Engle (2002) as examples of empirical implementations.

7 The interested reader can find more details in Barone-Adesi and Sala (2015).

8 For the Radon–Nikodym theorem to apply, the physical measure has to be σ-finite and atomless, while the risk-neutral measure has to be σ-finite and absolutely continuous with respect to the physical measure (Billingsley, 1995).

9 The focus of this paper is on the difference between $\mathcal{H}_t$ and $\mathcal{F}_t$, while we consider $\mathcal{F}_t = I_t$. Theoretically, by the fundamental theorems of asset pricing, $\mathcal{F}_t = I_t$ if the market is arbitrage-free and complete.

3. Data

In our empirical analysis we use option data provided by OptionMetrics. We use daily closing bid and ask prices on the S&P 500 Index options from January 2, 2002 to August 31, 2015. OptionMetrics also provides the S&P 500 daily (continuously compounded) dividend yield and the daily term-structure of risk-free zero-coupon interest rates. We only work with deep-out-of-the-money (DOTM) and out-of-the-money (OTM) European Put and Call options, as the in-the-money (ITM) options tend to be less liquid. We consider an option to be DOTM if k is larger than 1.15 (for Calls) or smaller than 0.85 (for Puts). Analogously, we consider an option to be OTM if k is between 1 and 1.15 (for Calls) and between 0.85 and 1 (for Puts). Following Barone-Adesi et al. (2008), we use the average of closing bid and ask prices, and we only keep options with prices larger than \$0.05, with implied volatility smaller than 70%, and with time-to-maturity τ between 10 and 360 days. On each day, and for each option maturity, we compute the relevant interest rate by linearly interpolating the term-structure of daily interest rates. As common in the option literature, we sample the data every Wednesday. In case a particular Wednesday is a holiday, we use the following trading day. The final dataset consists of 710 weekly option panels, for a total of 496,409 options, and an average of 700 OTM and DOTM option contracts per day, characterized by different combinations of moneyness and time-to-maturity. Finally, daily prices on the S&P 500 Index are obtained from CRSP. We use data from January 2, 1990 to August 31, 2015.

4. Estimating the empirical pricing kernel

In this section we describe how we estimate the *option-adjusted* EPK that we denote with $\text{EPK}^{\dagger}_{t,T}$, as opposed to the traditional EPK, that we simply denote with $\text{EPK}_{t,T}$. The difference between the two resides in the estimation of the conditional probability under the physical measure. We define the option-adjusted $\text{EPK}^{\dagger}_{t,T}$ as the EPK in which the denominator is computed by enriching the physical measure with information taken from the option market, while the $\text{EPK}_{t,T}$ is the traditional EPK, in which the physical probability at the denominator is solely based on historical returns. The construction of the $\text{EPK}^{\dagger}_{t,T}$, on each day t for a future time $T = t + \tau$, entails four distinct steps. First, we estimate the physical measures, $p_{t,T}$, using historical observations on the underlying stock (Section 4.1). Second, we estimate the risk-neutral measures, $q_{t,T}$, using option prices available on day t (Section 4.2). Third, we construct the new *option-adjusted* physical measure, $p^{\dagger}_{t,T}$, by combining the physical and risk-neutral measures through our proposed Bayesian approach (Sections 4.3 and 4.4). Fourth, we compute the *option-adjusted* $\text{EPK}^{\dagger}_{t,T}$, by taking the present value of the ratio between the risk-neutral measure computed in the second step and the option-adjusted physical measure computed in the third step (Section 4.5).

4.1. First step: Estimation of the physical measure

It is well established that daily S&P 500 Index returns exhibit stochastic volatility, negative skewness, and excess kurtosis. Volatility is persistent, mean-reverting, and responds asymmetrically to negative and positive returns (see, e.g.: Ghysels et al. (1996)). Volatility persistence is associated with extended periods of high and low volatility (i.e. so called volatility clustering). In order to capture these features of the data, we follow Barone-Adesi et al. (2008) and we employ an asymmetric GJR-GARCH model (see Glosten et al. (1993)) in which the shape of the returns distribution is estimated nonparametrically with the FHS (GJR-GARCH-FHS) of Barone-Adesi et al. (1999).

The choice of using the GJR-GARCH-FHS model is motivated by three facts. First, the GJR-GARCH (and the GJR-GARCH-FHS) model does not belong to the family of affine processes. It is well known that affine processes are not able to accurately capture the returns dynamics under the physical measure (see, for example, Bandi and Renò (2016)). Affine processes usually need to resort on multiple state variables (see Andersen et al. (2015)), whose identification, under the physical measure poses several challenges. Second, even if the GJR-GARCH-FHS model does not formally include any jump components, jumps (positive and negative) are implicitly modeled though the filtered historical innovations. For example Barone-Adesi et al. (2008) show that the GJR-GARCH-FHS is able to quantitatively and qualitatively match the high prices of OTM Put options on the S&P 500 Index. Finally, the GJR-GARCH and its extension, the GJR-GARCH-FHS, have been successfully used by Awartani and Corradi (2005) and Rosenberg and Engle (2002) (among others). Awartani and Corradi (2005) show that the GJR-GARCH is able to accurately forecast volatility over both short and long time horizons. Rosenberg and Engle (2002) study the EPK and find that GJR-GRACH-FHS, among the large family of GARCH models, is able to accurately describe the daily S&P 500 Index returns.

According to the GJR-GARCH-FHS model, the dynamics of the daily log-returns, under the physical measure, is given by:

$$\log\left(\frac{S_t}{S_{t-1}}\right) = \mu_t + \epsilon_t \tag{8}$$

$$\epsilon_t = \sqrt{\sigma_t^2} z_t \tag{9}$$

$$z_t \sim f_t(0, 1) \tag{10}$$

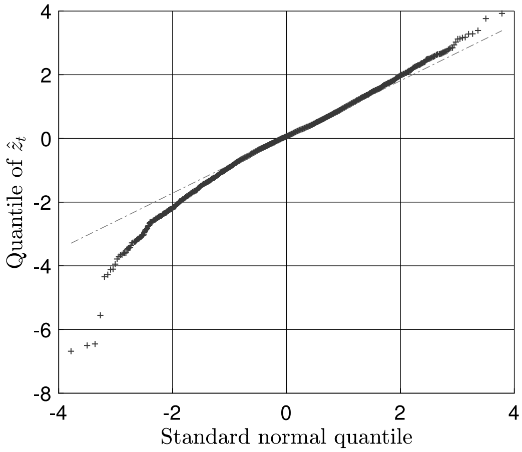

Fig. 2. QQ plot: Quantile–Quantile plot of the 2002–2015 empirical innovations estimated with Eq. (13).

$$\sigma_t^2 = \omega + \beta\sigma_{t-1}^2 + \alpha\epsilon_{t-1}^2 + \gamma \mathbb{1}_{i-1}\epsilon_{t-1}^2 \tag{11}$$

$$\mathbb{1}_{t-1} = \begin{cases} 1 & \text{if } \epsilon_{t-1} < 0, \\ 0 & \text{if } \epsilon_{t-1} \geq 0. \end{cases} \tag{12}$$

In Eq. (8), μ_t is the drift of the log-returns and it is equal to $\mu_t = (r_t - d_t - 0.5\sigma_t^2) + \phi_t$, where r_t is the risk-free rate, d_t is the dividend yield, $0.5\sigma_t^2$ is the usual convexity adjustment, and ϕ_t is the equity risk-premium. Following Merton (1980), Jackwerth (2000) and Barone-Adesi et al. (2008) (among others) we assume a constant equity risk-premium over the sample, and we set ϕ_t equal to 4% per year.[10] The shape of the returns distribution, defined in Eq. (10), is estimated nonparametrically. Specifically, the density function f_t is computed as the empirical distribution of past standardized return innovations $\hat{z}_t$. $\hat{z}_t$ is obtained by dividing past return innovations $\hat{\epsilon}_t$ by their conditional volatility:

$$\hat{z}_t = \frac{\hat{\epsilon}_t}{\sqrt{\sigma_t^2}} \tag{13}$$

where σ_t^2 is estimated according to Eq. (11). As for the variance dynamics in Eq. (11), ω, α, β, and γ control the long-run mean and the volatility persistence. γ is also responsible for generating the so called leverage-effect. The leverage-effect is the empirical negative correlation between returns and volatility. In the GJR-GARCH-FHS model, a negative return increases the variance of the next day by $\gamma\epsilon_{t-1}^2$.

Starting from January 2, 2002, we estimate the GJR-GARCH-FHS on each Wednesday using S&P 500 Index historical daily log-returns on an expanding time window starting from January 2, 1990.[11] Given that the model is re-estimated every week, the empirical distribution of $\hat{z}_t$ changes over time. In general, the distribution of $\hat{z}_t$ has mean zero, standard deviation equal to one, negative skewness, and positive excess kurtosis. Fig. 2 shows the QQ-plot of the distribution of the standardized return innovations computed over the entire sample. From Fig. 2 it is easy to note that the distribution of the filtered historical innovations is characterized by negative skewness (here equal to -0.47) and fat-tails (the excess kurtosis is here equal to 4.7). Table 1 (in the columns with header P) reports summary statistics of the estimated parameters $\theta_t = (\omega, \alpha, \beta, \gamma)$ obtained via Pseudo Maximum Likelihood (PML) (see Bollerslev and Wooldridge (1992)). The PML technique delivers consistent estimates even when f_t is non-normal (Gourieroux et al., 1984). The estimation is done weekly and Table 1 reports averages of the estimated parameters on each year. At each time t, given the estimated parameter vector θ_t, the transition density under the physical measure $p_{t,T}$ can be easily obtained via simulation.[12]

In Section 6, as a robustness check, we develop and estimate an extension of the GJR-GARCH-FHS that includes a stochastic long-run mean. We confirm that our results are robust to this different modeling choice.

4.2. Second step: Estimation of the risk-neutral measure

We assume that under the risk-neutral measure, the S&P 500 Index log-returns follow the same GJR-GARCH-FHS described in Section 4.1, but with a possibly different parameter vector $\widetilde{\theta}_t = (\widetilde{\omega}, \widetilde{\alpha}, \widetilde{\beta}, \widetilde{\gamma})$. Basically, the S&P 500 Index

[10] In unreported results, available upon request, we confirm that our findings are robust to alternative assumptions on the dynamics of ϕ_t. Fixing ϕ_t to 6% or allowing ϕ_t to change over time as in Duarte and Rosa (2015), for example, does not affect our results.

[11] We perform, every week, the Portmanteau test of Ljung and Box (1978) and the Lagrange multiplier (LM) ARCH test of Engle (1982). Both tests show that the model is effective in removing the volatility clustering observed in the S&P 500 Index returns. Results are omitted, but available upon request.

[12] Details on the simulation of the GJR-GARCH-FHS are provided in Section 4.2.

log-returns follow, under both the physical and risk-neutral measures, a stochastic process of the same family but with different parameters. The difference in parameters subsumes the change of measure, and does not place any restrictions on the shape of the PK. This differentiates our approach from part of the literature (e.g.: Rosenberg and Engle (2002), Bliss and Panigirtzoglou (2004) and Ross (2015)) that assume a parametric functional form for the PK.

On each Wednesday, the parameter vector $\widetilde{\theta}_t$ is estimated by minimizing the squared distance between observed ($\mathrm{O}^{\mathrm{obs}}_{t,k,\tau}$) and model-implied ($\mathrm{O}_{t,k,\tau}(\widetilde{\theta}_t, \sigma_t)$) option prices:

$$\min_{\widetilde{\theta}_t} \sum_K \sum_\tau \left(\mathrm{O}^{\mathrm{obs}}_{t,k,\tau} - \mathrm{O}_{t,k,\tau}(\widetilde{\theta}_t, \sigma_t)\right)^2, \tag{14}$$

where

$$\mathrm{O}_{t,k,\tau}(\widetilde{\theta}_t, \sigma_t) = \begin{cases} \mathrm{P}_{t,k,\tau}(\widetilde{\theta}_t, \sigma_t) & \text{if } k < 1, \\ \mathrm{C}_{t,k,\tau}(\widetilde{\theta}_t, \sigma_t) & \text{if } k \geq 1, \end{cases} \tag{15}$$

is the model-implied price of an OTM option with moneyness k and maturity τ. Since the GJR-GARCH-FHS does not belong to the family of affine process, option prices are not available in closed-from, and they have to be computed via simulation. Option prices are obtained by simulating N trajectories of the price process from t to $t + \tau$:

$$\log(\widetilde{S}^n_{t+i+1}) = \log(\widetilde{S}^n_{t+i}) + \widetilde{\mu}^n_i + \epsilon^n_{t+i}, \quad \text{with } n = 1 : N \text{ and } i = 1 : \tau, \tag{16}$$

where $\widetilde{\mu}^n_i = (r_t - q_t - (\widetilde{\sigma}^n_{t+i})^2/2)$ is the drift of the log-returns under the risk-neutral measure, while $\widetilde{S}^n_{t+i}$ and $\widetilde{\sigma}^n_{t+i}$ are the simulated price and volatility on day $t + i$ along the nth trajectory, under the risk-neutral measure. At each step and along each trajectory, the returns innovation $\widetilde{\epsilon}^n_{t+i}$ is simulated as:

$$\widetilde{\epsilon}^n_{t+i} = \widetilde{\sigma}^n_{t+i+1}\hat{z}, \tag{17}$$

where $\hat{z}$ is a random draw from the filtered historical innovations $\{\hat{z}_u\}_{\{u \leq t\}}$. Finally, the risk-neutral variance, at time $t + i$ and over the nth trajectory, follows a dynamics similar to Eq. (11), but with risk-neutral parameters ($\widetilde{\theta}_t$):

$$(\widetilde{\sigma}^n_{t+i+1})^2 = \widetilde{\omega} + \widetilde{\beta}(\widetilde{\sigma}^n_{t+i})^2 + \widetilde{\alpha}(\widetilde{\epsilon}^n_{t+i})^2 + \widetilde{\gamma}\mathbb{1}_i(\widetilde{\epsilon}^n_{t+i})^2. \tag{18}$$

We simulate a total of $N = 50{,}000$ sample paths[13] and we compute the price of European Call and Put options with maturity τ and strike price K as:

$$\mathrm{C}_{t,k,\tau}(\widetilde{\theta}_t, \sigma_t) = e^{-r_{t,\tau}\tau} \sum_{n=1}^{N} (\widetilde{S}^n_{t+\tau} - K, 0)^+ \tag{19}$$

$$\mathrm{P}_{t,k,\tau}(\widetilde{\theta}_t, \sigma_t) = e^{-r_{t,\tau}\tau} \sum_{n=1}^{N} (K - \widetilde{S}^n_{t+\tau}, 0)^+, \tag{20}$$

where $\widetilde{S}^n_{t+\tau}$ is the simulated price at the end of the nth path and $r_{t,\tau}$ is the relevant risk-free interest rate for a horizon equal to τ. Table 1 (in the columns with header Q) reports summary statistics of the estimated parameters $\widetilde{\theta}_t = (\widetilde{\omega}, \widetilde{\alpha}, \widetilde{\beta}, \widetilde{\gamma})$. At each time t, given the estimated parameter vector $\widetilde{\theta}_t$, the transition density under the risk-neutral measure $q_{t,T}$ can be easily obtained via simulation following the steps just explained.

It is worth noticing that, from the seminal work of Breeden and Litzenberger (1978), the risk-neutral measure can also be estimated nonparametrically (see the surveys of Bahara (1997) and Jackwerth (1999) and references therein). The advantage of using a parametric model, as we do in this section, is twofold. First, it allows not only to interpolate, but also to extrapolate option prices outside of the observed strike range. This allows to obtain smooth estimates of the risk-neutral measure far in the tails. Second, it produces, by construction, arbitrage free option prices. However, as a robustness check, in Section 6 we show that our results do not change when estimating the risk-neutral measure nonparametrically, as in Figlewski (2008).

4.3. Third step: Estimation of the option-adjusted physical measure

In this section we describe how to construct the option-adjusted physical measure $p^\dagger_{t,T}$. The end-goal is to provide a rigorous procedure that can be used to enrich the physical measure computed from historical returns with potentially forward-looking information extracted from the option market. We do so by following a Bayesian nonparametric approach by means of the Dirichlet process (DP). The DP is a stochastic process fully characterized by two parameters, α^* and C, and it is denoted DP(α^*, C). The base parameter, C, is a distribution representing the centering parameter as well as the mean

[13] We adopt the empirical martingale simulation (EMS) approach of Duan and Simonato (1998) to ensure that the simulated paths $\widetilde{S}^n_{t+\tau}$ have an expected rate of return equal $r_t - d_t$. The EMS approach ensures that the simulated paths do not violate the option pricing bounds.

of the DP. The parameter α^* is a positive scaling parameter.[14] Conceptually it is a measure of precision that indicates how much importance is given to the prior information. Higher (lower) values of α^* give more (less) weight to the prior information encoded in C. In our case, the prior information is the option market information. Following a nonparametric approach, both the prior and the resulting posterior are allowed to belong to the space of all admissible distributions. As such, the DP does not restrict any distribution to pertain to any specific parametric family. In summary: the DP is a nonparametric approach that avoids the critical dependence on parametric assumptions while exploiting, through the prior, the information coming from an external source, here the option market. Appendix A contains technical details on the DP.

Let $p_{t,T}$ and $q_{t,T}$ be the physical and the risk-neutral measure, respectively. Since we want to enrich the physical measure with the information coming from the risk-neutral measure, we need to modify the risk-neutral measure such that its expected value reflects the one of the physical measure. We define the resulting measure the *option-based* physical measure, $\widetilde{p}_{t,T}$. The new measure $\widetilde{p}_{t,T}$ has the same shape (i.e. central moments higher than one) of the risk-neutral measure, $q_{t,T}$, but its first moment is equal to the mean of $p_{t,T}$, which reflects the equity risk premium.[15] We assume that $\widetilde{p}_{t,T}$ has a DP distribution. This prior distribution will be updated given the new information coming from the underlying (from which we compute the physical measure $p_{t,T}$). This is done by exploiting the conjugacy of the DP. The DP is a conjugate family of priors over distributions, that is closed under posterior updates given observations. Being the DP a collection of stable distributions it follows that, after observing the new market realization, the one time updated DP posterior is still a DP with posterior mean (see Appendix A):

$$p^{\dagger}_{t,T} = \left(\frac{\alpha^*_t}{(\alpha^*_t+1)}\widetilde{p}_{t,T} + \frac{1}{(\alpha^*_t+1)}p_{t,T}\right) \tag{21}$$

$$= w_t\widetilde{p}_{t,T} + (1-w_t)p_{t,T}. \tag{22}$$

We call $p^{\dagger}_{t,T}$ the *option-adjusted* physical measure. $p^{\dagger}_{t,T}$ is a weighted average of the *option-based* physical measure and the physical measure, where the weights, w_t and $(1-w_t)$ are determined by the time-varying precision parameter α^*_t. In particular, we have the following limits:

$$\lim_{\alpha^*_t\to 0} p^{\dagger}_{t,T} = p_{t,T} \qquad \text{and} \qquad \lim_{\alpha^*_t\to +\infty} p^{\dagger}_{t,T} = \widetilde{p}_{t,T}. \tag{23}$$

That is, when α^*_t approaches zero, $p^{\dagger}_{t,T}$ converges to the physical measure computed in Section 4.1, as the weight given to the option information is zero. On the contrary, when α^*_t approaches infinity, $p^{\dagger}_{t,T}$ converges towards the option-based physical measure $\widetilde{p}_{t,T}$. For positive values of w_t the option-adjusted measure $p^{\dagger}_t$ has mean equal to the equity risk premium, but its higher moments are a mixture of the moments of the physical and risk-neutral distributions, with w_t determining the mixing intensity. In the next section we describe how to determine, on day-by-day basis, the value of α^*_t.

4.4. Linking the precision parameter to the option market

This section describes how we link the physical measure to the option market through the DP precision parameter α^*_t. It is well-known that options are more profitable when volatility is high. French and Roll (1980) show that stock market volatility is related to private and public information, and it tends to increase with the trading activity of informed traders. Mayhew et al. (1999), among others, find that option market liquidity increases with volatility and Chakravarty et al. (2004) show that option informativeness (defined as option market price discovery) is positively related to trading activity and stock market volatility. Finally, Wei and Zheng (2010) argue that there is an option moneyness-substitution effect induced by stock return volatility. They show that, when the stock return volatility increases, trading activity shifts from ITM options towards DOTM options. Given the evidence above, we link α^*_t to the availability of DOTM options. Specifically, we set the weights w_t (thus determining the value of α^*_t through Eqs. (21)–(22)) equal to the proportion of listed DOTM options over the total number of listed OTM and DOTM options on each single day. Fig. 3 shows the time series of the percentage of listed DOTM options (top panel), the percentage of DOTM option volume, (mid panel) and the VIX index (bottom panel) over the period 2002–2015.[16] As can be seen from Fig. 3, the number of listed DOTM options correlates significantly with the trading volume and, more in general, with the level of volatility (proxied by the VIX index). Computing w_t as a function of the ratio between the daily volume of DOTM options over the total volume of OTM and DOTM volume, or computing w_t as a function of the VIX index, leads to very similar results.[17]

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[14] The parameter α^* takes different names in the literature e.g.: the centering, the precision or the strength parameter. Throughout the paper we will refer to it as the precision parameter. While in literature the precision parameter is usually identified with α, we use α^* to prevent confusion with the GARCH parameter α.

[15] The risk premium adjustment is needed for consistency with respect to the physical measure.

[16] For example, for the first day in our sample, 02/01/2002, the percentage of listed DOTM is 48.69%. It follows that $w_t = 0.4869$ and $(1-w_t) = 0.5131$. Also, given that $w_t = \alpha^*_t/(1-\alpha^*_t)$ it is trivial to obtain the value of α^*_t given w_t.

[17] The correlation between the percentage of DOTM options listed and the percentage of DOTM option volume is 0.83, while the correlation between the percentage of DOTM options listed and the VIX index is equal to 0.63.

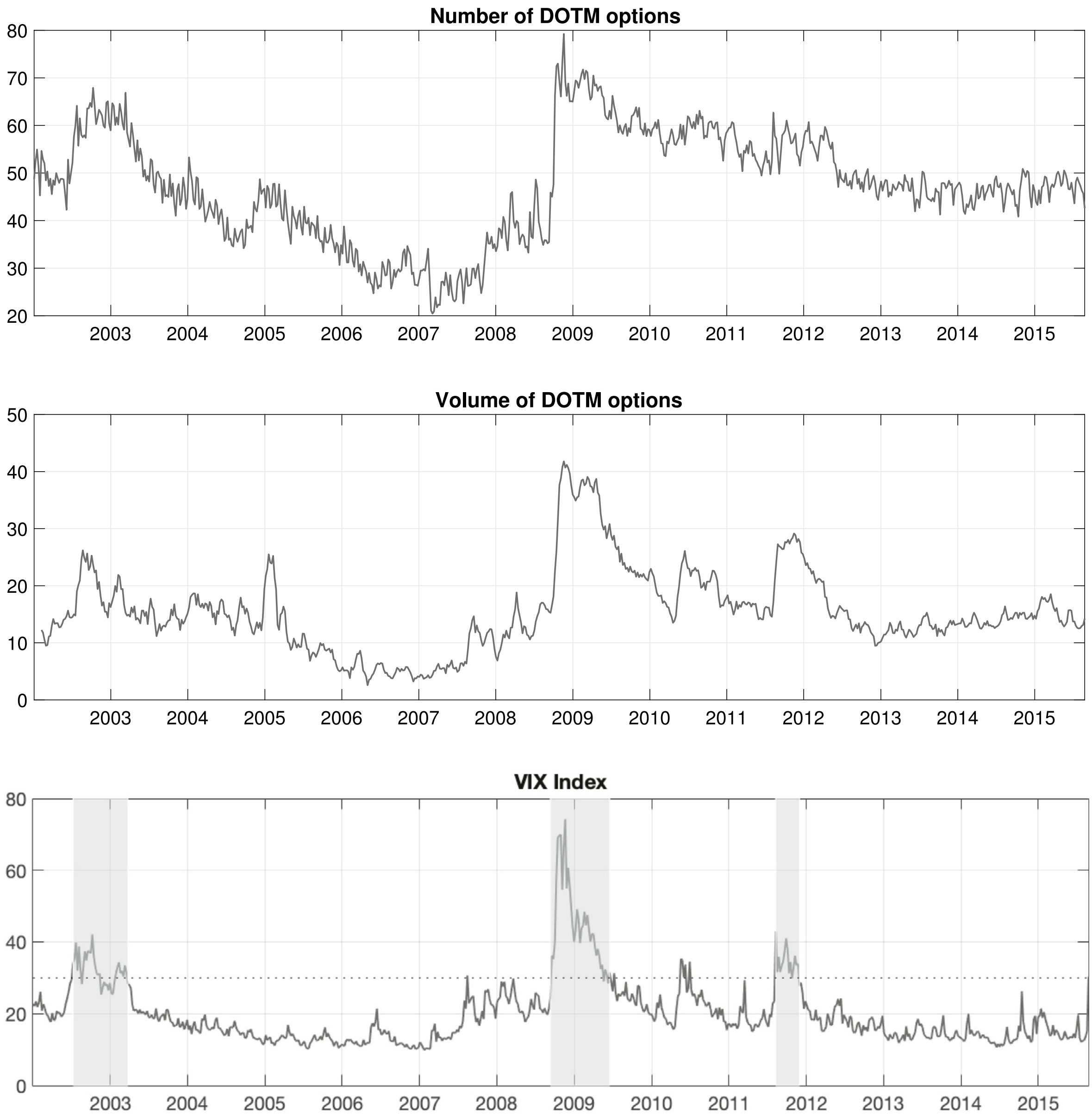

Fig. 3. DOTM options and volatility: The top panel shows the time series of the percentage of listed DOTM options over the total number of listed DOTM and OTM options. The middle panel shows the time series of the percentage of the volume of DOTM option over the total DOTM and OTM volume. The bottom panel shows the value of the CBOE VIX index. The shaded areas correspond to periods in which the VIX Index is higher than 30%.

Finally, following the Bayesian approach, it would be possible to place a higher level in the modeling hierarchy by eliciting a prior also on the α_t^* parameter, following West (1995). Also, a more flexible distribution than the DP could be utilized. One possible choice is given by the Pitman and Yor (1997) process, which through the discount parameters, allows for a better modeling of the tails of the distribution. We leave the study of these alternative modeling choices for future research.

4.5. Fourth step: Estimation of the option-adjusted pricing kernel

As defined in Eq. (1), the estimation of the PK requires the estimation of a risk-neutral measure at the numerator and a physical measure at denominator. In our approach we use the risk-neutral measure, $q_{t,T}$, computed according to Section 4.2 and the option-adjusted physical measure, $p_{t,T}^{\dagger}$, defined by Eq. (21) in Section 4.3. It follows that we need to compute three measures, the risk-neutral measure, $q_{t,T}$, the physical measure, $p_{t,T}$, and the option-adjusted physical measure, $p^{\dagger}_{t,T}$, which is the weighted average of $p_{t,T}$ and $\widetilde{p}_{t,T}$.

The three conditional probabilities are obtained by simulation, similar to what is done in Section 4.2. To compute the physical measure, $p_{t,T}$, we simulate from the model in Eq. (8) by using the estimated parameter vector θ_t and setting the log-return drift equal to $\mu_t = (r_t - d_t - \sigma_t^2/2) + \phi_t$. The risk-neutral measure, $q_{t,T}$, is computed analogously, but using the estimated parameter vector $\widetilde{\theta}_t$ and a drift equal to $\widetilde{\mu}_t = (r_t - d_t - \widetilde{\sigma}_t^2/2)$. Finally, the option-based physical measure, $\widetilde{p}_{t,T}$, is obtained by simulating form the parameter vector $\widetilde{\theta}_t$, as in the risk-neutral measure, but by using a drift equal to $\mu_t = (r_t - d_t - \sigma_t^2/2) + \phi_t$, as in the physical measure. We employ the Empirical Martingale Simulation (EMS) approach of Duan and Simonato (1998) to make sure that the average value of the simulated returns trajectories is exactly equal to the theoretical drift. Finally, the different densities are inferred from the simulated log-returns using a Gaussian kernel evaluated at equidistant points with optimal bandwidth computed following the Silverman's rule of thumb. We ensure that all densities are nonnegative and integrate to one. Lastly, following Eq. (22) the option-adjusted $\mathrm{PK}^{\dagger}_{t,T}$ is equal to:

$$M^{\dagger}_{t,T} = e^{-r_t\cdot\tau}\left(\frac{q_{t,T}}{w_t\cdot\widetilde{p}_{t,T} + (1-w_t)\cdot p_{t,T}}\right) = e^{-r_t\cdot\tau}\left(\frac{q_{t,T}}{p^{\dagger}_{t,T}}\right) \tag{24}$$

It is worth noticing that the traditional $\mathrm{PK}_{t,T}$ of Barone-Adesi et al. (2008) is a special case of Eq. (24) when α_t^* tends to zero:

$$M_{t,T} = \lim_{\alpha_t^*\to 0} M^{\dagger}_{t,T} = e^{-r_t\cdot\tau}\left(\frac{q_{t,T}}{p_{t,T}}\right) \tag{25}$$

The other limit, when α_t^* tends to infinity (or equivalently when w_t tends to one), is more complicated. In this case we have:

$$M_{t,T} = \lim_{\alpha_t^*\to +\infty} M^{\dagger}_{t,T} = e^{-r_t\cdot\tau}\left(\frac{q_{t,T}}{\widetilde{p}_{t,T}}\right) \tag{26}$$

If the equity risk-premium is assumed to be zero, then the resulting PK would be a flat line. A positive risk premium, by shifting the distribution at the denominator to the right, would generate a monotonically decreasing PK, thus mechanically resolving the pricing kernel puzzle. A central point of our procedure is that we do not impose any a-priori structure on the weights w_t, but rather we let the trading activity in the option market to determine how large the probability adjustment (i.e. w_t) has to be, and how w_t changes over time. Empirically, as can be seen from Fig. 3, the average value of the weight w_t is slightly below 0.5, with its lowest and highest values reaching 0.2 and 0.8, respectively, thus remaining quite far from the limiting case of one. Eqs. (24) and (25) refer to the theoretical version of the PK. In the Empirical results section, we investigate the properties of the option-adjusted $\mathrm{EPK}^{\dagger}_{t,T}$ ($\widehat{M}^{\dagger}_{t,T}$) and the traditional $\mathrm{EPK}_{t,T}$ ($\widehat{M}_{t,T}$).

5. Empirical results

In this section we provide a description of the shape of the option-adjusted and traditional EPKs. The traditional EPK corresponds to the method of Barone-Adesi et al. (2008), which is used as a benchmark in our analysis. We estimate the $\mathrm{EPK}^{\dagger}_{t,T}$ and $\mathrm{EPK}_{t,T}$ on each Wednesday from January 2, 2002 to August 31, 2015. On each day, $\tau = T - t$ corresponds to the different options' maturities available. We end up with 710 daily estimates. In Section 5.1 we present summary results discussing the unconditional estimates of the two EPKs, while in Section 5.2 we analyze the shape of the EPKs on specific days.

5.1. Summary estimates

In discussing the behavior of the EPK it is useful to distinguish between periods of high and low volatility. In the existing literature, U-shaped EPKs are mostly observed in high-volatility periods. According to the evidence presented in Section 4.4, it is precisely in high volatility periods that the trading activity in DOTM options increases, and this might make the option market informative about the future dynamics of the underlying asset. This intuition is at the core of the construction of the $\mathrm{EPK}^{\dagger}_{t,T}$. As a consequence, we expect the $\mathrm{EPK}^{\dagger}_{t,T}$ and the $\mathrm{EPK}_{t,T}$ to behave quite differently when volatility is elevated.

We divide our sample in two sub-samples. We define low-volatility-days as the days in which the VIX index is below its 2002–2015 sample median (around 18%) and high-volatility-days as the days in which the VIX is above its median. Within each period, we then compute the average EPK using a univariate Gaussian kernel with bandwidths selected according to the Silverman's rule of thumb. The results are reported in Fig. 4. The dark line and the light gray line show the option-adjusted and the traditional EPKs, from the model in Eq. (8). The error bands correspond to the 90% confidence intervals. From Fig. 4 we can draw two conclusions. First, on days of low volatility (left panel), the traditional and the option-adjusted EPKs deliver substantially similar results. It is also evident that the hump-shape of the EPKs near the at-the-money region remains visible. Second, in periods of high volatility (right panel), while the traditional $\mathrm{EPK}_{t,T}$ is U-shaped, the option-adjusted $\mathrm{EPK}^{\dagger}_{t,T}$ remains monotonically decreasing, as predicted by economic theory.

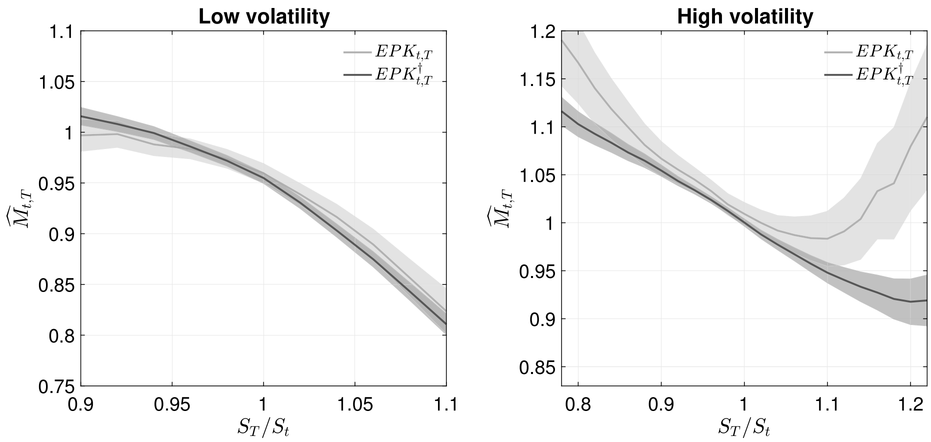

Fig. 4. Empirical pricing kernel from GJR-GARCH-FHS model: The figure shows the option-adjusted $\mathrm{EPK}^{\dagger}_{t,T}$ (dark line) and the traditional $\mathrm{EPK}_{t,T}$ (light gray line) along with their 90% confidence intervals. The left (right) panel corresponds to low (high) volatility days (i.e. VIX index smaller (larger) than its median value over the sample). The figure is constructed by averaging the daily estimated EPK using a univariate Gaussian kernel with bandwidth selected according to the Silverman's rule of thumb.

5.2. Daily estimates

In this section we show the behavior of the EPK on specific days in the sample. Different from the unconditional analysis presented in the previous section, by looking at a specific day, we can more directly study the shape of the EPK as a function of the time horizon $\tau = T - t$, and can relate this to the market characteristics on that day. Again, we focus on high and low volatility days. From observing the dynamics of the VIX index reported at the bottom of Fig. 3, we see that there are three periods that are characterized by elevated volatility, in which the VIX index exceeds 30%. The three periods are highlighted, in Fig. 3, by a shaded gray area. The first period, which coincides with the 2002 market crash, goes from July 2002 to March 2003; the second period, marked by the 2008 housing bubble and financial crisis, starts in September 2008 and ends in June 2009; and the third period, which corresponds to the second European crisis, goes from August 2011, to November 2011. In this section we analyze one day taken from each one of the three high volatility periods and one day taken from a low volatility day. We report additional results in Appendix D.

Fig. 5 shows the option-adjusted and traditional EPKs for January 14, 2009, a day characterized by high volatility. For this day, the VIX index is equal to 49.14%, almost three times the VIX sample median. The figure shows, in the left panels, the physical measure $p_{t,t+\tau}$ (dashed line), the risk-neutral measure $q_{t,t+\tau}$ (dotted line), and the *option-adjusted* physical measure $p^{\dagger}_{t,t+\tau}$ (continuous line), respectively. The right panels show the option-adjusted EPK, $\widehat{M}^{\dagger}_{t,t+\tau}$, (continuous line) and the traditional EPK, $\widehat{M}_{t,t+\tau}$ (dotted line). The different rows correspond to different time horizons, which coincide with the available options maturities on that day. In all cases we notice that the option-adjusted physical measures $\mathrm{p}^{\dagger}_{t,t+\tau}$ lie between the $p_{t,t+\tau}$ and $q_{t,t+\tau}$. The impact of $\mathrm{p}^{\dagger}_{t,t+\tau}$ on the shape of the EPK is clear. While the $\mathrm{EPK}_{t,t+\tau}$ is characterized by a marked U-shape, the $\mathrm{EPK}^{\dagger}_{t,t+\tau}$ maintains a monotonically decreasing pattern for all maturities. The day just analyzed is characterized by a high level of volatility as well by a high level of listed and traded DOTM options. These market characteristics confirm what postulated by Cuesdeanu and Jackwerth (2017). In their broad survey, they trace the origin of the EPK puzzle(s) to two possible factors: (1) a large presence of DOTM Call options and (2) the use of a backward-looking physical measure. The evidence provided in this section points toward the fact that option market data, and specifically the information embedded into DOTM options, can improve the estimation of the physical measure and leads to monotonically decreasing EPKs, as implied by economic theory.

Next, we consider a day characterized by low volatility and a low percentage of DOTM listed and traded: March 20, 2002. The results are reported in Fig. 6. Different from before, the two EPKs are now almost on top of each other. This confirms what we reported, unconditionally, in the left panel of Fig. 4. In periods of low volatility the traditional and the option-adjusted EPKs deliver extremely similar results.

6. Alternative estimation approaches

In order to assess the validity of our approach we perform two different robustness checks. First, we show that our results are not determined by the particular model used to describe the return dynamics. To this end, we develop and estimate a two-factor extension of the GJR-GARCH-FHS used in Section 4.1. Second, we show that our results are not due to a possibly miss-specified model under the risk-neutral measure. We address this issue by estimating the risk-neutral measure nonparametrically.

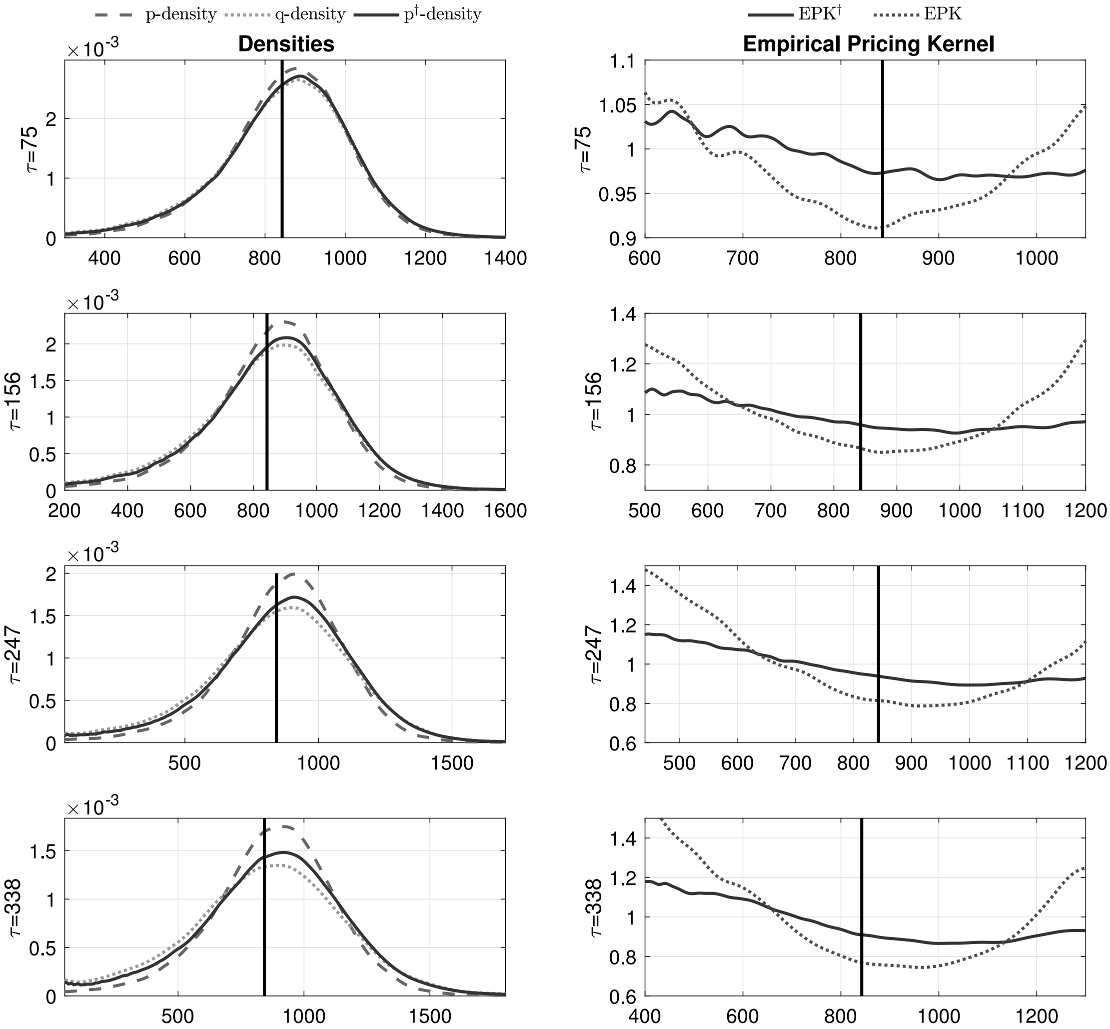

Fig. 5. Daily estimation results on January 14, 2009: The left panels show the estimated physical ($p_{t,\tau}$, dashed line), risk-neutral ($q_{t,\tau}$, dotted line), and option-adjusted physical ($\widetilde{p}_{t,\tau}$, continuous line) measures. The right panels show the corresponding traditional (EPK, dotted line) and option-adjusted (EPK†, continuous line) EPKs. The different rows represent the EPKs over different time horizons, which correspond to the available options maturities.

6.1. C-GJR-GARCH-FHS

In this section we develop a two-factor model to describe the dynamics of the S&P 500 Index. The model is similar to the two-factor GARCH model in Engle and Lee (1999) and the Component GARCH in Christoffersen et al. (2008). Different from the aforementioned studies we rely on the FHS to describe the distribution of the empirical innovations, instead of assuming normal innovations. The two factors should help capturing transient and persistent volatility components, and this should translate into more precise estimates of both physical and risk-neutral measures over short and long time horizons. We label this model Component GJR-GARCH with Filtered Historical Simulation, or simply C-GJR-GARCH-FHS. Specifically, according to the C-GJR-GARCH-FHS, the dynamics of the log-returns, under the physical measure, is given by:

$$\log\left(\frac{S_t}{S_{t-1}}\right) = \mu_t + \epsilon_t \tag{27}$$

$$\sigma_t^2 = h_t + \alpha_s(\epsilon_{t-1}^2 - h_{t-1}) + \beta_s(\sigma_{t-1}^2 - h_{t-1}) + \gamma_s(\mathbb{1}_{t-1}\epsilon_{t-1}^2 - \frac{1}{2}h_{t-1}) \tag{28}$$

$$h_t = \omega + \alpha_l\epsilon_{t-1}^2 + \beta_l h_t + \gamma_l\mathbb{1}_{t-1}\epsilon_{t-1}^2 \tag{29}$$

$$\mathbb{1}_{t-1} = \begin{cases} 1, & \text{if } \epsilon_{t-1} < 0 \\ 0 & \text{if } \epsilon_{t-1} \geq 0, \end{cases} \tag{30}$$

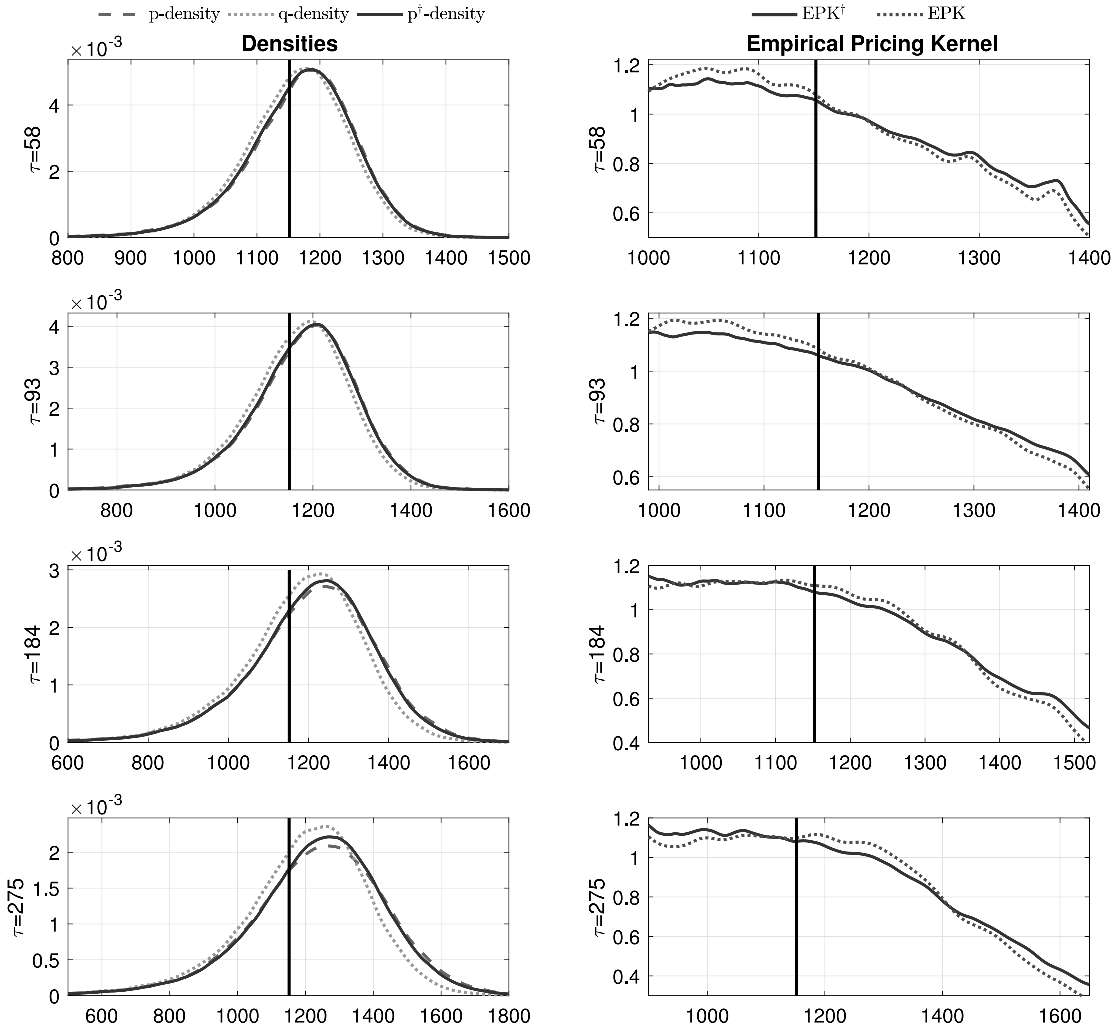

Fig. 6. Daily estimation results on March 20, 2002: The left panels show the estimated physical ($p_{t,\tau}$, dashed line), risk-neutral ($q_{t,\tau}$, dotted line), and option-adjusted physical ($\widetilde{p}_{t,\tau}$, continuous line) measures. The right panels show the corresponding traditional (EPK, dotted line) and option-adjusted (EPK†, continuous line) EPKs. The different rows represent the EPKs over different time horizons, which correspond to the available options maturities.

where h_t is the time-varying long-run mean (i.e. the persistent component) of σ_t^2, and $(\sigma_t^2 - h_t)$ is the short-run (i.e. the transient) component of σ_t^2. By construction, the unconditional mean of the short-run component $(\sigma_t^2 - h_t)$ is equal to zero. Fig. 7 shows the volatility and the persistent volatility component estimated over the entire sample. It is evident that h_t captures volatility fluctuation along the business-cycle, while the transient component captures momentary spikes in volatility. We estimate the physical and risk-neutral measures following the same procedures described in Sections 4.1 and 4.2. The physical parameters of the C-GJR-GARCH-FHS model $\theta_c = (\omega, \alpha_s, \beta_s, \gamma_s, \alpha_l, \beta_l, \gamma_l)$ and their risk-neutral counterparts, $\widetilde{\theta}_c = (\widetilde{\omega}, \widetilde{\alpha}_s, \widetilde{\beta}_s, \widetilde{\gamma}_s, \widetilde{\alpha}_l, \widetilde{\beta}_l, \widetilde{\gamma}_l)$, are summarized in Table 2 (in the columns with header P and Q, respectively). We then construct the PK according to the C-GJR-GARCH-FHS model. We denote with $M_{t,T}^C$ the PK constructed according to the traditional (component) method, and with $M_{t,T}^{\dagger C}$ the PK constructed using the (component) option-adjusted probability measure. Specifically:

$$M_{t,T}^{\dagger C} = e^{-r_t \cdot \tau}\left(\frac{q_{t,T}^C}{w_t \cdot \widetilde{p}_{t,T}^C + (1-w_t)\cdot p_{t,T}^C}\right) = e^{-r_t \cdot \tau}\left(\frac{q_{t,T}^C}{p_{t,T}^{\dagger C}}\right), \tag{31}$$

where $q_{t,T}^C$, $p_{t,T}^C$, and $\widetilde{p}_{t,T}^C$ are the risk-neutral, the traditional physical, and the option-adjusted physical measures based on the C-GJR-GARCH-FHS parameters. It is worth noticing that the traditional (component) $\text{PK}_{t,T}^C$ is a special case of Eq. (31)

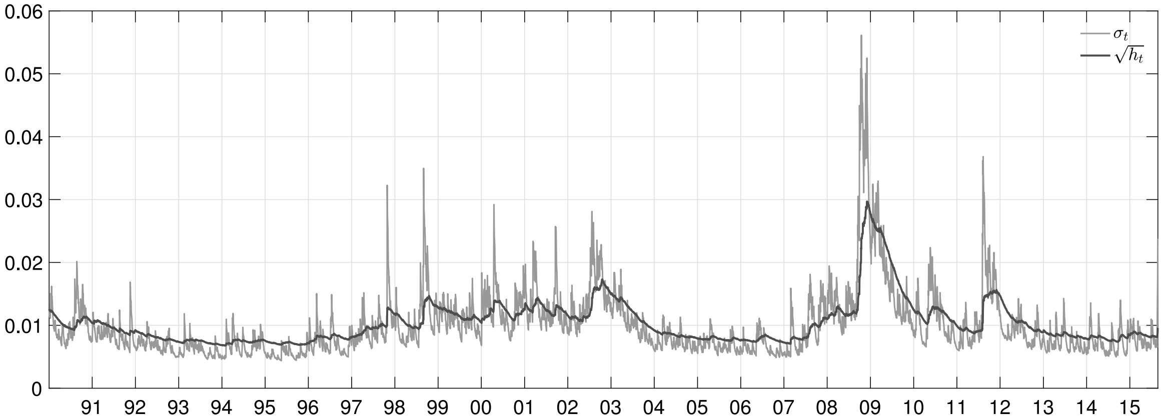

Fig. 7. C-GJR-GARCH-FHS: Time-series of the short and long-run volatility estimated by the CGJR-GARCH model in Eq. (27).

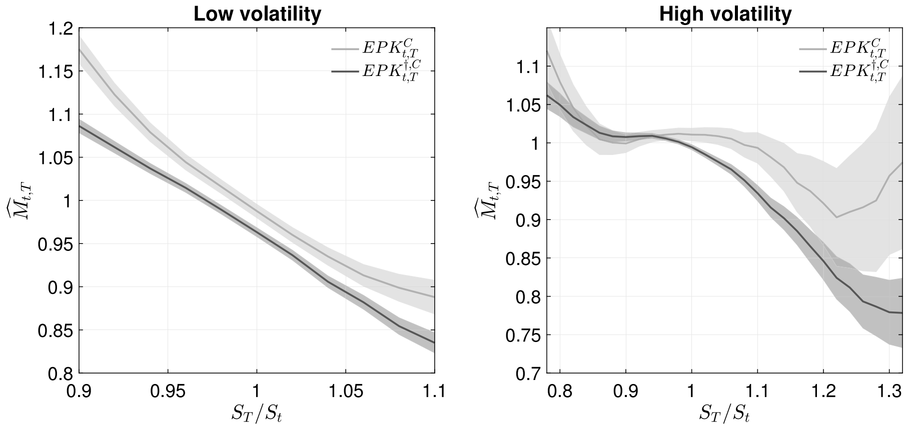

Fig. 8. Empirical pricing kernel from C-GJR-GARCH-FHS model: The figure shows the average option-adjusted $\text{EPK}^{\dagger,C}_{t,T}$ (dark line) and the traditional (component) $\text{EPK}^{C}_{t,T}$ (light gray line) along with their 90% confidence intervals. The left (right) panel corresponds to low (high) volatility days (i.e. VIX index smaller (larger) than its median value over the sample). The figure is constructed by averaging the daily estimated EPKs using a univariate Gaussian kernel with bandwidth selected according to the Silverman's rule of thumb.

when α^*_t tends to zero:

$$M^C_{t,T} = \lim_{\alpha^*_t \to 0} M^{\dagger C}_{t,T} = e^{-r_t \cdot \tau} \left(\frac{q_{t,T}}{p_{t,T}} \right) \tag{32}$$

We investigate the shape of the empirical versions of Eqs. (31) and (32) defined as $\text{EPK}^{\dagger C}_{t,T}$ ($\widehat{M}^{\dagger C}_{t,T}$) and $\text{EPK}^{C}_{t,T}$ ($\widehat{M}^{C}_{t,T}$), respectively. Fig. 8 shows the unconditional shape of the $\text{EPK}^{\dagger C}_{t,T}$ (dark line) and $\text{EPK}^{C}_{t,T}$ (light gray line) in low and high volatility periods. Fig. 8 shows that: (1) the U-shape of $\text{EPK}^{C}_{t,T}$ is still present in high-volatility periods, even when adding a second volatility factor; (2) the option-adjusted $\text{EPK}^{\dagger C}_{t,T}$ is monotonically decreasing; (3) the addition of a second volatility factor seems to attenuate the hump-shape of both the EPKs in low volatility periods.

6.2. Nonparametric risk-neutral measure

Capturing the dynamics of the risk-neutral measure can be a challenge for a parametric model. In order to control for possible misspecification of our model in pricing options, we follow the seminal work of Breeden and Litzenberger (1978) and estimate the risk-neutral measure nonparametrically using a procedure similar to Figlewski (2008). Specifically, on each day, and given a specific option maturity, the nonparametric risk-neutral measure $q^{NP}_{t,T}$ is estimated by taking the second derivative of the Call option prices with respect to the strike price:

$$q^{NP}_{t,T}(K) = e^{r_t \tau} \frac{d^2 C_{t,k,\tau}}{dK^2} \tag{33}$$

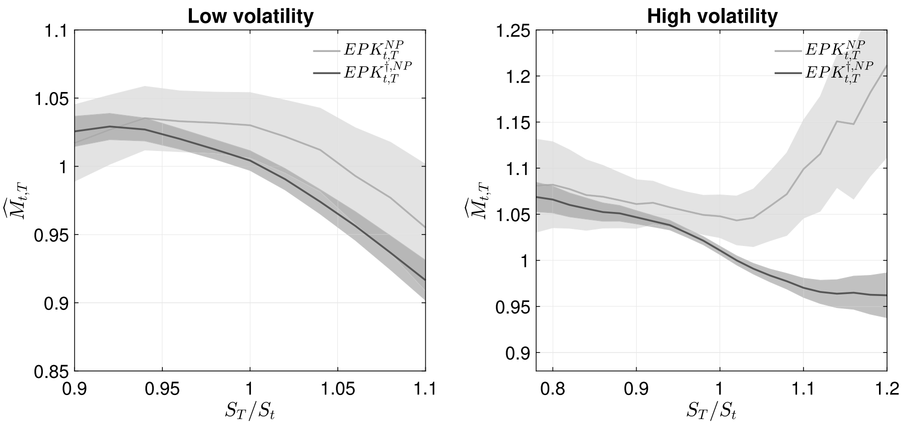

Fig. 9. Empirical pricing kernel from GJR-GARCH-FHS model with nonparametric risk-neutral measure: The figure shows the average option-adjusted $\text{EPK}_{t,T}^{\dagger,NP}$ (dark line) and the traditional (non-parametric) $\text{EPK}_{t,T}^{NP}$ (light gray line) along with their 90% confidence intervals. The left (right) panel corresponds to low (high) volatility days (i.e. VIX index smaller (larger) than its median value over the sample). The figure is constructed by averaging the daily estimated EPKs using a univariate Gaussian kernel with bandwidth selected according to the Silverman's rule of thumb.

where $C_{t,k,\tau}$ is the price of a European Call option at time t, with maturity τ and moneyness k and K is the option strike price. Since taking the second derivative of noisy option market prices is very often an ill-posed problem (curse of differentiation) we first fit a second degree polynomial on the option implied volatility (as a function of the strike prices), and then take the second derivative of the smoothed implied volatility. Finally, we construct the (non-parametric) option-adjusted PK, $\text{M}_{t,T}^{\dagger NP}$ as:

$$M_{t,T}^{\dagger NP} = e^{-r_t \cdot \tau} \left(\frac{q_{t,T}^{NP}}{w_t \cdot \widetilde{p}_{t,T} + (1 - w_t) \cdot p_{t,T}} \right) = e^{-r_t \cdot \tau} \left(\frac{q_{t,T}^{NP}}{p_{t,T}^{\dagger}} \right), \tag{34}$$

where the physical measure is the same as the one estimated in Section 4.1. Once more, the traditional (non-parametric) $\text{PK}_{t,T}^{NP}$ is a special case of Eq. (34) when α_t^* tends to zero:

$$M_{t,T}^{RN} = \lim_{\alpha_t^* \to 0} M_{t,T}^{\dagger RN} = e^{-r_t \cdot \tau} \left(\frac{q_{t,T}}{p_{t,T}} \right) \tag{35}$$

We investigate the shape of the empirical versions of Eqs. (34) and (35) defined as $\text{EPK}_{t,T}^{\dagger RN}$ ($\widehat{M}_{t,T}^{\dagger RN}$) and $\text{EPK}_{t,T}^{RN}$ ($\widehat{M}_{t,T}^{RN}$), respectively. Fig. 9 confirms that our results do not change when using $q_{t,T}^{NP}$ in the computation of the EPK. While the traditional method still shows a pronounced U-shape in the high volatility regime, our method produces a monotonically decreasing EPK.

7. Conclusion

According to economic theory the PK should be monotonically decreasing in wealth. However, a large body of literature finds that the EPK is U-shaped, and that the U-shape is particularly pronounced in periods of elevated volatility. We argue that the U-shape can be due to an inconsistency between the estimated risk-neutral and physical measures that enter in the numerator and the denominator of the EPK. While the risk-neutral measure is inherently *forward-looking*, the estimation of the physical measure requires the use of historical data, making it a *backward-looking* measure. This informational mismatch becomes more severe in high volatility periods, when economic conditions change rapidly, and historical data might be less informative about the future economic conditions. To alleviate this informational mismatch, we develop a new *option-adjusted* physical measure that combines information from past returns and the option market. We then use this new option-adjusted physical measure in the construction of an *option-adjusted* pricing kernel. An extensive empirical analysis on the S&P 500 Index shows that the newly proposed pricing kernel is monotonically decreasing, as opposed to the traditional pricing kernel which is characterized by a pronounced U-shape in high volatility periods.

Acknowledgments

We are grateful to two anonymous referees and to Yacine Äit-Sahalia (the Editor) for comments and suggestions that greatly improved the paper. We would also like to thank Loriano Mancini, Antonio Mele, Eckhard Platen, Francesco Franzoni, Alberto Plazzi, Patrick Gagliardini, Hersh Shefrin, Jerôme Detemple, David Bates, Stijn Van Nieuwerburgh, Fabio

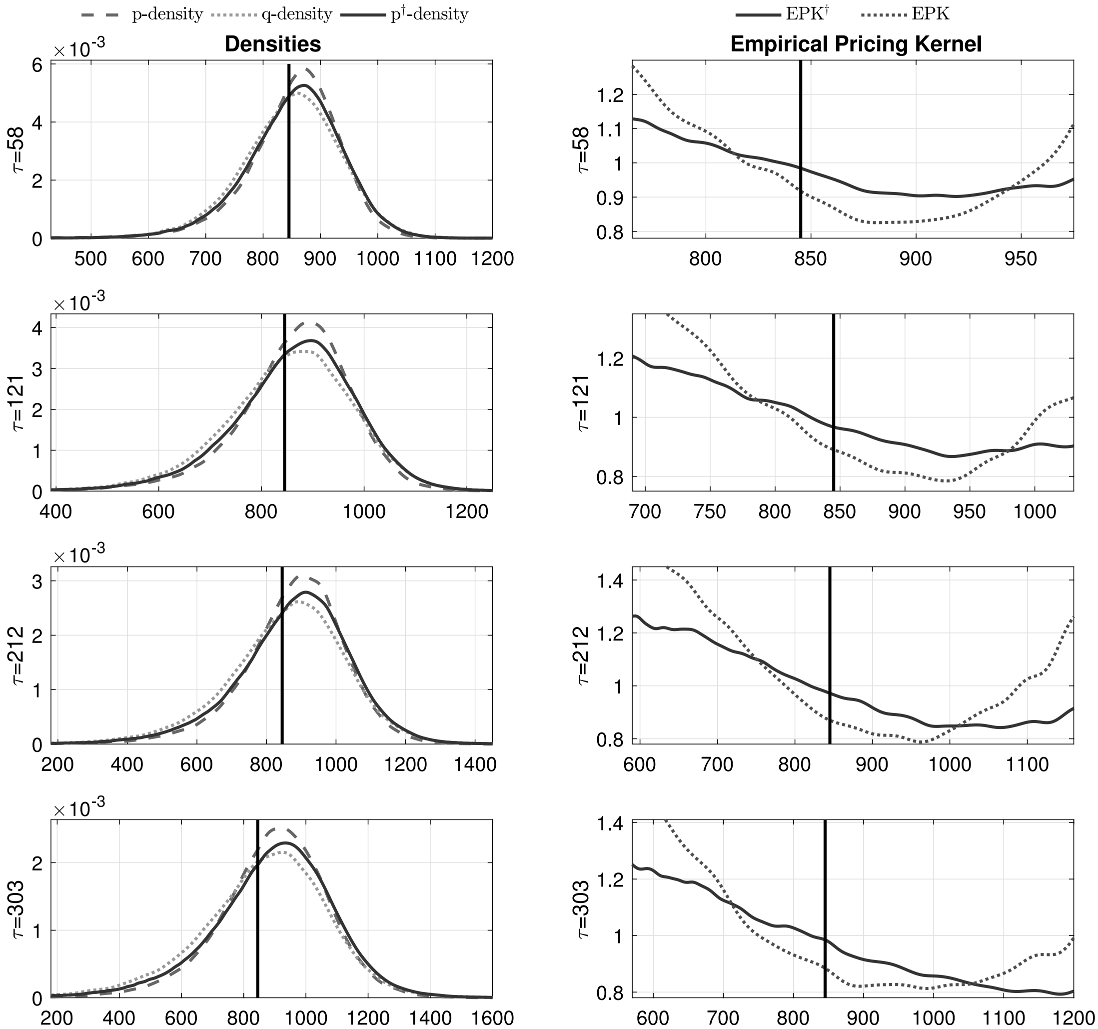

Fig. 10. Daily estimation results on February 19, 2003: The left panels show the estimated physical ($p_{t,\tau}$, dashed line), risk-neutral ($q_{t,\tau}$, dotted line), and option-adjusted physical ($\widetilde{p}_{t,\tau}$, continuous line) measures. The right panels show the corresponding traditional (EPK, dotted line) and option-adjusted (EPK†, continuous line) EPKs. The different rows represent the EPKs over different time horizons, which correspond to the available options maturities.

Bellini, Zorana Grbac, and Antonio Rubia and seminar and conference participants at the University of Lugano, University of Lausanne, ESADE Business School, University of Milano Bicocca, 2015 SoFIE Summer School in Brussels, XIV Conference in Quantitative Finance in Parma, SFI Research Day, Istanbul BISP9 conference, EWGCFM conference in Milan, the Barcelona XXV Finance Forum and the Workshop on Parameter Uncertainty with Focus on Recovery for their valuable comments. This work was supported by the SNF 100018-172892 grant and AGAUR - SGR 2017-640.

Appendix A. The Dirichlet Process (DP)

Introduced by Ferguson (1973), the DP is a distribution over distributions. It is called DP because it is a stochastic process whose samples are discrete probability measures (generically referred as Random Probability Measures (RPM)) such that its marginals with respect to any finite partition has Dirichlet distribution.[18] Let us consider a probability space $(\Omega, \mathcal{A}, c)$ and an arbitrary partition $A_1, A_2, \ldots, A_n$ of Ω. Denoted by $c \sim DP(\alpha^*, C)$, the random distribution c is said to follow a DP with precision parameter α^* and base distribution C if:

$$(c(A_1), c(A_2), \ldots, c(A_n)) \sim \mathrm{Dir}(\alpha^* C(A_1), \alpha^* C(A_2), \ldots, \alpha^* C(A_n)) \tag{A.1}$$

18 The Dirichlet distribution is a multivariate generalization of the Beta distribution.

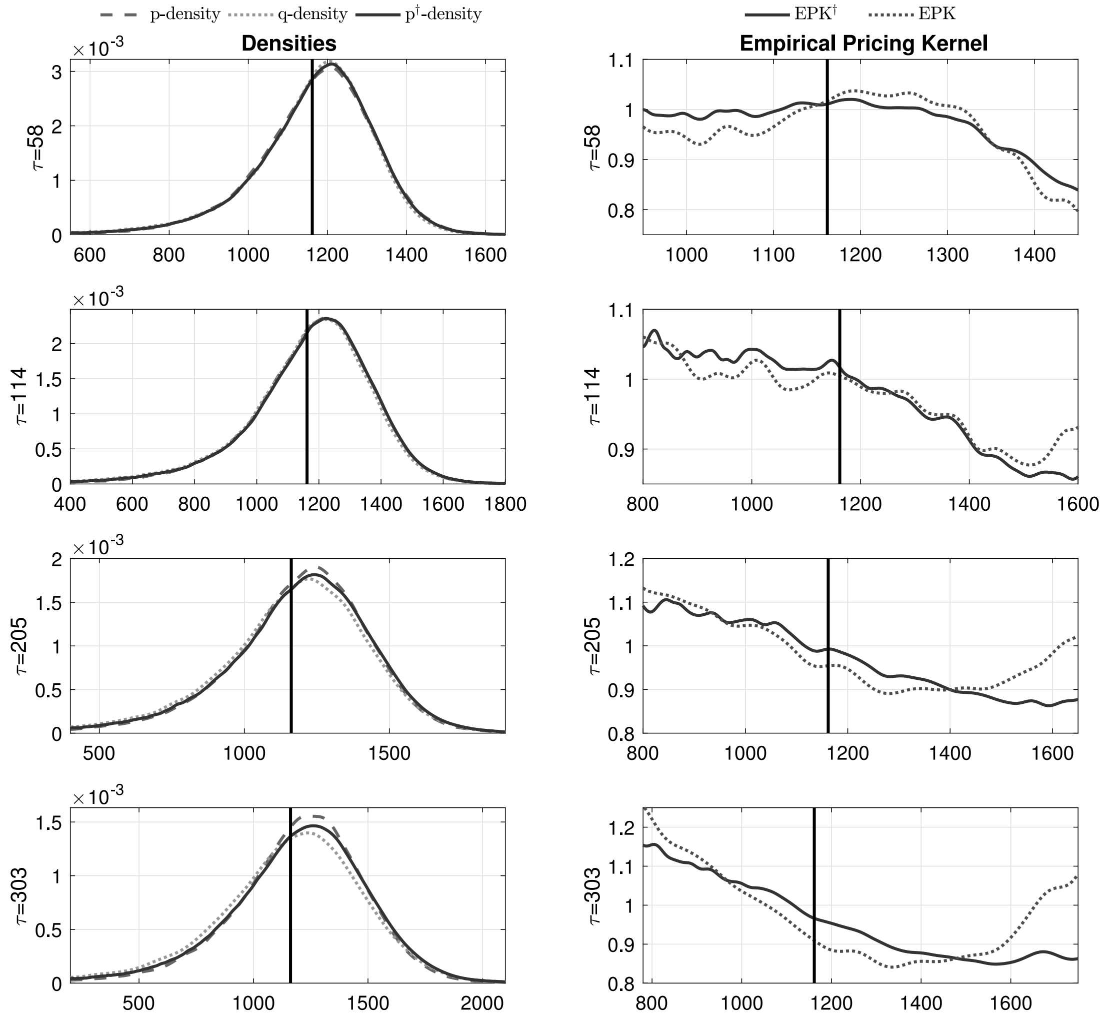

Fig. 11. Daily estimation results on November 23, 2011: The left panels show the estimated physical ($p_{t,\tau}$, dashed line), risk-neutral ($q_{t,\tau}$, dotted line), and option-adjusted physical ($\widetilde{p}_{t,\tau}$, continuous line) measures. The right panels show the corresponding traditional (EPK, dotted line) and option-adjusted (EPK†, continuous line) EPKs. The different rows represent the EPKs over different time horizons, which correspond to the available options maturities.

As demonstrated by Ferguson (1973) the collection of finite dimensional distributions in Eq. (A.1) satisfies the Kolmogorov consistency condition and implies a well defined infinite dimensional model.

It follows that, by placing a distribution on the random measure c, then for any $A \subset \Omega$, the quantity $c(A)$ is a random variable. Ferguson (1973) shows that $c(A) \sim \text{Beta}\ \{\alpha^* C(A), \alpha^*(1 - C(A))\}$, such that:

$$\mathbb{E}[c(A)] = C(A), \qquad \mathbb{V}[c(A)] = \frac{C(A)(1 - C(A))}{\alpha^* + 1} \tag{A.2}$$

which shows that the base parameter C is the expected shape of the random distribution c (i.e. its mean) and the precision parameter α^* affects the variability of the realizations around the base distribution C. Among its properties, the DP is a conjugate family of priors over distributions that is closed under posterior updates given observations. Assume p_i for $i = 1, \ldots, n$ to be an i.i.d sample of observations with $p_i | c \sim c$ and $c \sim DP(\alpha^*, C)$, then:

$$c | p_1, \ldots, p_n \sim DP\left(\alpha^* + n, \frac{\alpha^* C \sum_{i=1}^{n} \delta_{p_i}}{\alpha^* + n}\right) \tag{A.3}$$

which, for a single update ($n = 1$):

$$c | p_1 \sim DP\left(\alpha^* + 1, \frac{\alpha^* C \sum_{i=1}^{1} \delta_{p_i}}{\alpha^* + 1}\right) \tag{A.4}$$

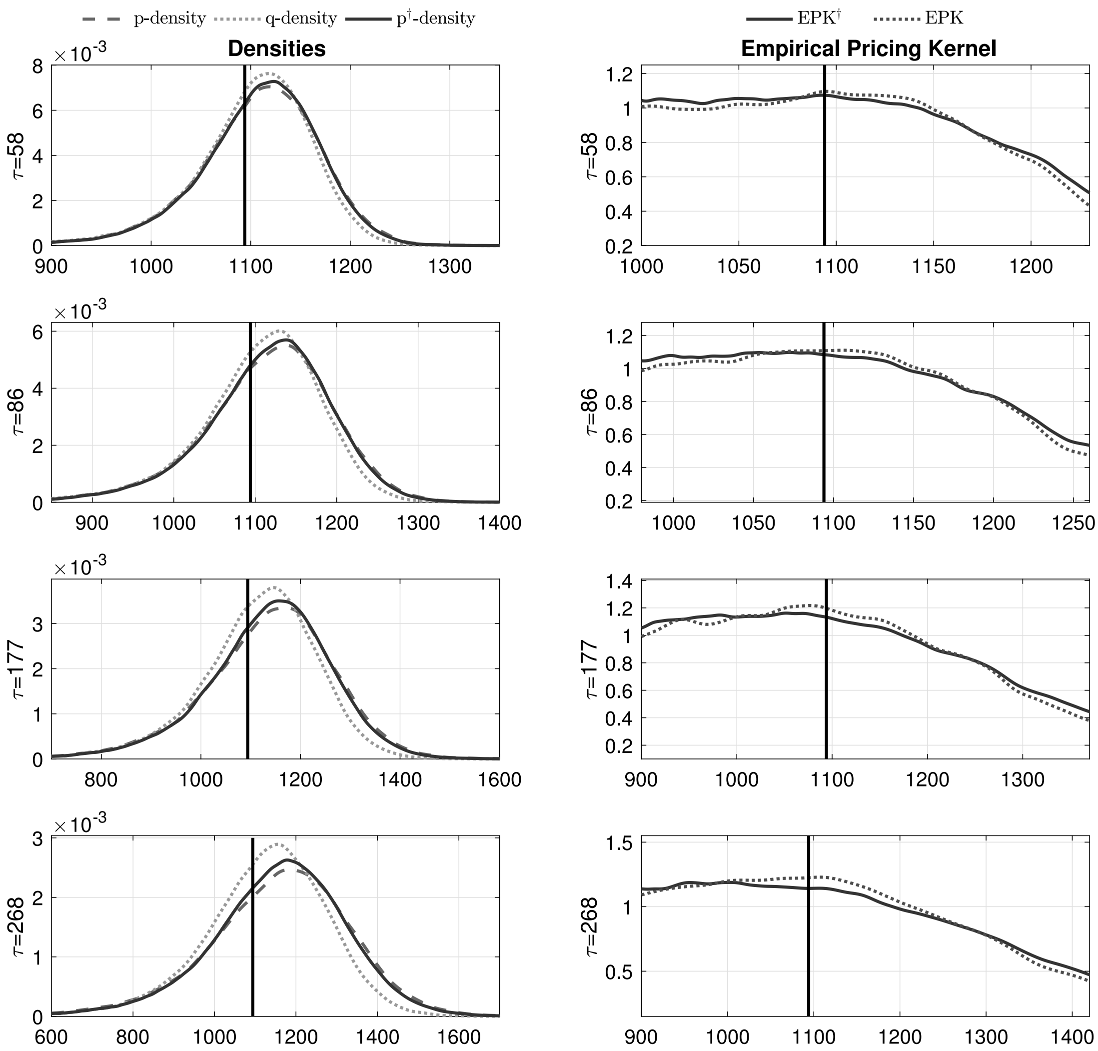

Fig. 12. Daily estimation results on December 24, 2003: The left panels show the estimated physical ($p_{t,\tau}$, dashed line), risk-neutral ($q_{t,\tau}$, dotted line), and option-adjusted physical ($\widetilde{p}_{t,\tau}$, continuous line) measures. The right panels show the corresponding traditional (EPK, dotted line) and option-adjusted (EPK†, continuous line) EPKs. The different rows represent the EPKs over different time horizons, which correspond to the available options maturities.

It follows that the posterior mean is given by:

$$\mathbb{E}(c|p_1,\ldots,p_n) = \frac{\alpha^* C \sum_{i=1}^{n} \delta_{p_i}}{\alpha^* + n} \tag{A.5}$$

which, for a single update (n = 1), is equal to:

$$\begin{aligned}\mathbb{E}(c|p_1) &= \frac{\alpha^* C \sum_{i=1}^{1} \delta_{p_i}}{\alpha^* + 1} \\ &= \frac{\alpha^*}{\alpha^* + 1} C + \frac{1}{\alpha^* + 1} p_1\end{aligned} \tag{A.6}$$

In our setting, Eq. (A.6) is equivalent to Eq. (21) in the main text, where the base parameter, C, is the daily option-based physical measure, $\widetilde{p}_{t,T}$, p_1 is the daily traditional physical measure, $p_{t,T}$, and α^* corresponds to $\alpha^*_{t,T}$ as defined in Section 4.3. Finally, even though the DP is an almost surely discrete RPM, a continuous distribution can be easily obtained via kernel smoothing.

Table 1
Summary statistics of the physical (P) and risk-neutral (Q) parameters for the GJR-GARCH-FHS model in Eq. (8). The table reports the yearly average of the parameters estimated on each Wednesday, from January 2, 2002 to August 31, 2015, on the S&P 500 Index.

GJR	ω		α		β		γ	
Year	P	Q	P	Q	P	Q	P	Q
2002	1.34e−06	2.20e−06	0.010	0.005	0.918	0.890	0.118	0.172
2003	1.24e−06	1.79e−06	0.009	0.005	0.922	0.903	0.116	0.154
2004	1.21e−06	7.11e−07	0.008	0.001	0.924	0.920	0.112	0.142
2005	1.14e−06	1.71e−06	0.008	0.006	0.926	0.815	0.108	0.313
2006	1.10e−06	1.01e−06	0.007	0.003	0.927	0.873	0.108	0.214
2007	1.16e−06	1.63e−06	0.005	0.001	0.927	0.845	0.110	0.275
2008	1.17e−06	7.39e−06	0.003	0.001	0.928	0.819	0.112	0.306
2009	1.17e−06	4.75e−06	0.002	0.001	0.928	0.868	0.116	0.230
2010	1.23e−06	2.51e−06	0.002	0.002	0.926	0.862	0.119	0.250
2011	1.30e−06	2.95e−06	0.001	0.001	0.925	0.850	0.122	0.276
2012	1.44e−06	1.72e−06	0.000	0.001	0.921	0.866	0.131	0.250
2013	1.49e−06	1.49e−06	7.06e−07	0.000	0.920	0.856	0.133	0.259
2014	1.53e−06	1.89e−06	5.38e−05	0.001	0.917	0.800	0.136	0.363
2015	1.62e−06	1.03e−06	7.09e−06	2.37e−05	0.913	0.858	0.143	0.266

Notes: The parameters under the physical measure (P) are estimated via Pseudo Maximum Likelihood (PML) as described in Section 4.1. The parameters under the risk-neutral measure (Q) are computed by calibrating the GJR-GARCH-FHS to the available OTM options on the S&P 500 Index. The calibration is performed by minimizing the sum of squared pricing errors between the observed and the model implied option prices, as described in Section 4.2.

Table 2
Summary statistics of the physical (P) and risk-neutral (Q) parameters for the C-GJR-GARCH-FHS model in Eq. (27). The table reports the yearly average of the parameters estimated on each Wednesday, from January 2, 2002 to August 31, 2015, on the S&P 500 Index.

GJR	ω		α_s		β_s		γ_s		α_l		β_l		γ_l	
Year	P	Q	P	Q	P	Q	P	Q	P	Q	P	Q	P	Q
2002	1.39e−07	5.87e−05	1.13e−05	0.010	0.841	0.738	0.165	0.376	0.007	0.011	0.991	0.916	5.19e−07	0.074
2003	1.55e−07	7.01e−06	9.59e−06	0.012	0.850	0.742	0.159	0.325	0.007	0.002	0.991	0.863	1.11e−06	0.156
2004	2.34e−07	7.46e−07	2.93e−06	0.001	0.846	0.788	0.156	0.241	0.009	0.001	0.987	0.969	7.12e−08	0.048
2005	2.25e−07	8.81e−05	1.22e−07	0.003	0.848	0.739	0.151	0.398	0.009	0.009	0.987	0.948	5.13e−07	0.045
2006	2.07e−07	2.72e−04	5.69e−08	0.010	0.852	0.785	0.149	0.328	0.009	0.008	0.988	0.969	1.69e−07	0.016
2007	2.15e−07	1.22e−04	1.08e−07	0.004	0.858	0.770	0.146	0.316	0.008	0.018	0.989	0.873	5.85e−07	0.135
2008	2.28e−07	4.18e−05	6.94e−08	0.001	0.866	0.718	0.142	0.374	0.008	0.002	0.989	0.928	1.26e−06	0.089
2009	2.69e−07	1.29e−04	6.68e−08	0.001	0.874	0.706	0.142	0.463	0.008	8.232	0.989	0.808	1.73e−07	0.219
2010	3.93e−07	3.36e−04	2.02e−07	0.002	0.872	0.782	0.144	0.284	0.009	0.016	0.986	0.887	9.02e−08	0.137
2011	4.05e−07	1.09e−04	6.37e−07	3.850	0.872	0.756	0.146	0.322	0.009	0.006	0.986	0.891	2.14e−07	0.140
2012	4.37e−07	5.45e−05	7.17e−07	4.920	0.878	0.810	0.149	0.276	0.008	0.002	0.987	0.913	1.51e−07	0.112
2013	4.37e−07	6.08e−05	1.87e−06	0.002	0.875	0.777	0.152	0.353	0.008	0.008	0.987	0.934	3.34e−07	0.065
2014	4.22e−07	4.14e−04	1.73e−06	0.009	0.868	0.736	0.158	0.433	0.008	0.022	0.987	0.937	3.80e−07	0.045
2015	4.12e−07	1.87e−04	2.44e−06	0.001	0.860	0.756	0.167	0.385	0.008	0.028	0.987	0.897	5.39e−07	0.092

Notes: The parameters under the physical measure (P) are estimated via Pseudo Maximum Likelihood (PML) as described in Section 4.1. The parameters under the risk-neutral measure (Q) are computed by calibrating the C-GJR-GARCH-FHS to the available OTM options on the S&P 500 Index. The calibration is performed by minimizing the sum of squared pricing errors between the observed and the model implied option prices, as described in Section 4.2.

Appendix B. GJR-GARCH-FHS estimation results

See Table 1.

Appendix C. C-GJR-GARCH-FHS estimation results

See Table 2.

Appendix D. Additional empirical results

In this section we report additional results on the shape of the EPKs in high and low volatility days. Specifically, Figs. 10 and 11 show the EPKs on two days in high volatility periods (see gray areas in the bottom panel of Fig. 3): February 19, 2003 and November 23, 2011. In contrast, Fig. 12 reports results for December 24, 2003, a day with low volatility.

References

Äit-Sahalia, Y., Lo, A.W., 1998. Non-parametric estimation of state-price densities implicit in financial asset prices. J. Finance 53 (2), 499–547.
Andersen, T.G., Fusari, N., Todorov, V., 2015. The risk premia embedded in index options. J. Financ. Econ. 117 (3), 558–584.
Awartani, B., Corradi, V., 2005. Predicting the volatility of the S&P-500 stock index via GARCH models: the role of asymmetries. Int. J. Forecast. 21, 167–183.
Bahara, B., 1997. Implied risk-neutral probability density functions from options prices: theory and application. Bank of England ISSN 1368-5562.
Bandi, F., Renò, R., 2016. Price and volatility co-jumps. J. Financ. Econ. 119, 107–146.
Barone-Adesi, G., Engle, R., Mancini, L., 2008. A GARCH option pricing model with Filtered Historical Simulation. Rev. Financ. Stud. 21, 1223–1258.
Barone-Adesi, G., Giannopoulos, K., Vosper, L., 1999. Backtesting derivative portfolios with Filtered Historical Simulation (FHS). J. Future Mark. 8 (1), 31–58.
Barone-Adesi, G., Sala, C., 2015. Sentiment Lost: the Effect of Projecting the Empirical Pricing Kernel onto a Smaller Filtration Set. Swiss Finance Institute Research Paper No. 15-58.
Barone-Adesi, G., Sala, C., 2019. Testing market efficiency with the pricing kernel. Eur. J. Financ. 1, 1–28.
Beare, B.K., 2011. Measure preserving derivatives and the pricing kernel puzzle. J. Math. Econom. 47, 689–697.
Beare, B.K., Schmidt, D.W.L., 2016. An empirical test of pricing kernel monotonicity. J. Appl. Econometrics 1099–1255.
Billingsley, P., 1995. Probability and Measure, third ed. Wiley.
Bliss, R.R., Panigirtzoglou, N., 2004. Option-implied risk aversion estimates. J. Finance 59, 407–446.
Bollerslev, T., Wooldridge, J.M., 1992. Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariances. Econometric Rev. 11 (2), 143–172.
Borovicka, J., Hansen, L., Scheinkman, J., 2016. Misspecified recovery. J. Finance 71 (6), 2493–2544.
Breeden, D., Litzenberger, R., 1978. Prices of state-contingent claims implicit in option prices. J. Bus. 51 (4), 621–651.
Chabi-Yo, F., 2012. Pricing kernels with stochastic skewness and volatility risk. Manage. Sci. 58 (3), 624–640.
Chabi-Yo, F., Renault, E., Garcia, R., 2007. State dependence can explain risk-aversion puzzle. Rev. Financ. Stud. 21 (2), 973–1011.
Chakravarty, S., Gulen, H., Mayhew, S., 2004. Informed trading in stock and option markets. J. Finance 59 (3), 1235–1257.
Christoffersen, P., Heston, S., Jacobs, K., 2013. Capturing option anomalies with a variance-dependent pricing kernel. Rev. Financ. Stud. 26, 1963–2006.
Christoffersen, P., Jacobs, K., Ornthanalai, C., Wang, Y., 2008. Option valuation with long-run and short-run volatility components. J. Financ. Econ. 90 (3), 272–297.
Cochrane, J.H., 2001. Asset Pricing. Princeton University Press.
Constantinides, G.M., Perrakis, S., 2002. Stochastic dominance bounds on derivatives prices in a multiperiod economy with proportional transaction costs. J. Econom. Dynam. Control 26, 1323–1352..
Cuesdeanu, H., Jackwerth, J., 2017. The pricing kernel puzzle: Survey and outlook. Ann. Financ..
Duan, J.C., Simonato, J.G., 1998. Empirical martingale simulation for asset prices. Manag. Sci. 44 9, 1218–1233.
Duarte, F., Rosa, C., 2015. The Equity Risk Premium: A Review of Models. Federal Reserve Bank of New York Staff Reports, (714).
Dybvig, P.H., 1988. Distributional analysis of portfolio choice. J. Bus. 61 (3), 369–393.
Engle, R., 1982. Autoregressive conditional heteroscedasticity with estimates of the variance of united kingdom inflation. Econometrica 50 (4), 987–1007.
Engle, R., Lee, G., 1999. A Permanent and Transitory Component Model of Stock Return Volatility. Oxford University Press.
Ferguson, T., 1973. Bayesian analysis of some nonparametric problems. Ann. Statist. 1 (2), 209–230.
Figlewski, S., 2008. Estimating the implied risk neutral density for the U.S. market portfolio. In: Bollerslev, Tim, Russell, Jeffrey R., Watson, Mark (Eds.), Volume Volatility and Time Series Econometrics: Essays in honor of Robert F. Engle. Oxford University Press.
French, K.R., Roll, R., 1980. Stock return variance: The arrival of information and the reaction of traders. J. Financ. Econ. 17, 5–26.
Ghysels, E., Harvey, A.C., Renault, E., 1996. Stochastic volatility. Handbook of Statist. 14, 119–191.
Glosten, L.R., Jagannathan, R., Runkle, D.E., 1993. On the relation between the expected value and the volatility of the nominal excess return on stocks. J. Finance 48 (5), 1779–1801.
Gourieroux, C., Monfort, A., Trognon, A., 1984. Pseudo maximum likelihood methods: Theory. Econometrica 52 (3), 681–700.
Grith, M., Härdle, W., Kratschmer, V., 2017. Reference-dependent preferences and the empirical pricing kernel puzzle. Rev. Financ. 21 (1), 269–298.
Jackwerth, J., 1999. Option-implied risk-neutral distributions and implied binomial trees: A literature review. J. Deriv. 7 (2), 66–82.
Jackwerth, J., 2000. Recovering risk aversion from option prices and realized returns. Rev. Financ. Stud. 13, 433–451.
Levy, H., 1985. Upper and lower bounds of put and call option value: Stochastic dominance approach. J. Finance 40, 1197–1217.
Linn, M., Shive, S., Shumway, T., 2017. Pricing kernel monotonicity and conditional information. Rev. Financ. Stud. 31 (2), 493–531.
Liu, X., Shackleton, M.B., Taylor, S.J., Xu, X., 2009. Empirical pricing kernels obtained from the UK index options market. Appl. Econ. Lett. 16 (10), 989–993.
Ljung, G.M., Box, G.E.P., 1978. On a measure of lack of fit in time series of models. Biometrika 65, 297–303.
Mayhew, S., Sarin, A., Shastri, K., 1999. What Drives Option Liquidity? Working paper.
Merton, R.C., 1980. On estimating the expected return on the market. J. Financ. Econ. 8, 323–361.
Perrakis, S., Ryan, P.J., 1984. Option pricing bounds in discrete time. J. Finance 39 (2), 519–525.
Pitman, J., Yor, M., 1997. The two-parameter Poisson-Dirichlet distribution derived from a stable subordinator. Ann. Probab. 25 (2), 855–900.
Rosenberg, V., Engle, R., 2002. Empirical pricing kernels. Rev. Financ. Stud. 64 (341–372).
Ross, S., 1976. Options and efficiency. Q. J. Econ. 90, 74–89.
Ross, S., 2015. The recovery theorem. J. Finance 70 (2), 615–648.
Rubinstein, M., 1994. The jump-risk premia implicit in options: Evidence from an integrated time-series study. J. Finance 49 (3), 771–818.
Song, Z., Xiu, D., 2016. A tale of two option markets: Pricing kernels and volatility risk. J. Econometrics 190, 176–196.
Wei, J., Zheng, J., 2010. Trading activity and bid–ask spreads of individual equity options. J. Bank. Financ. 34 (12), 2897–2916.
West, M., 1995. Hyperparameter Estimation in Dirichlet Process Mixture Models. Technical Report, Duke University.
Ziegler, A., 2007. Why does implied risk aversion smile? Rev. Financ. Stud. 20 (3), 859–904.